

9

Morphological Image Processing

In form and feature, face and limb,
I grew so like my brother
That folks got taking me for him
And each for one another.

Henry Sambrook Leigh, Carols of Cockayne, The Twins

Preview

The word *morphology* commonly denotes a branch of biology that deals with the form and structure of animals and plants. We use the same word here in the context of *mathematical morphology* as a tool for extracting image components that are useful in the representation and description of region shape, such as boundaries, skeletons, and the convex hull. We are interested also in morphological techniques for pre- or postprocessing, such as morphological filtering, thinning, and pruning.

In the following sections, we will develop a number of fundamental concepts in mathematical morphology, and illustrate how they are applied in image processing. The material in this chapter begins a transition from methods whose inputs and outputs are images, to methods whose outputs are image attributes, for tasks such as object extraction and description. Morphology is one of several tools developed in the remainder of the book—such as segmentation, feature extraction, and object recognition—that form the foundation of techniques for extracting “meaning” from an image. The material in the following sections of this chapter deals with methods for processing both binary and grayscale images.

Upon completion of this chapter, readers should:

- Understand basic concepts of mathematical morphology, and how to apply them to digital image processing.
- Be familiar with the tools used for binary image morphology, including erosion, dilation, opening, closing, and how to combine them to generate more complex tools.
- Be able to develop algorithms based on binary image morphology for performing tasks such as morphological smoothing, edge detection, extracting connected components, and skeletonizing.
- Be familiar with how binary image morphology can be extended to grayscale images.
- Be able to develop algorithms for grayscale image processing for tasks such as textural segmentation, granulometry, computing grayscale image gradients, and others.

Before proceeding, you will find it helpful to review the discussion in Section 2.4 dealing with representing images, the discussion on connectivity in Section 2.5, and the discussion on sets in Section 2.6.

9.1 PRELIMINARIES

The language of mathematical morphology is set theory. As such, morphology offers a unified and powerful approach to numerous image processing problems. When working with images, sets in mathematical morphology represent objects in those images. In binary images, the sets in question are members of the 2-D integer space Z^2 , where each element of a set is a tuple (2-D vector) whose coordinates are the coordinates of an object (typically foreground) pixel in the image. Grayscale digital images can be represented as sets whose components are in Z^3 . In this case, two components of each element of the set refer to the coordinates of a pixel, and the third corresponds to its discrete intensity value. Sets in higher dimensional spaces can contain other image attributes, such as color and time-varying components.

Morphological operations are defined in terms of sets. In image processing, we use morphology with two types of sets of pixels: *objects* and *structuring elements* (SE's). Typically, objects are defined as sets of foreground pixels. Structuring elements can be specified in terms of both foreground and background pixels. In addition, structuring elements sometimes contain so-called “don't care” elements, denoted by $\times$, signifying that the value of that particular element in the SE does not matter. In this sense, the value can be ignored, or it can be made to fit a desired value in the evaluation of an expression; for example, it might take on the value of a pixel in an image in applications in which value matching is the objective.

Because the images with which we work are rectangular arrays, and sets in general are of arbitrary shape, applications of morphology in image processing require that sets be embedded in rectangular arrays. In forming such arrays, we assign a background value to all pixels that are not members of object sets. The top row in Fig. 9.1 shows an example. On the left are sets in the graphical format you are accustomed to seeing in book figures. In the center, the sets have been embedded in a rectangular background (white) to form a graphical image.[†] On the right, we show a digital image (notice the grid) which is the format we use for digital image processing.

Structuring elements are defined in the same manner, and the second row in Fig. 9.1 shows an example. There is an important difference between the way we represent digital images and digital structuring elements. Observe on the top right that there is a border of background pixels surrounding the objects, while there is none in the SE. As you will learn shortly, structuring elements are used in a form similar to spatial convolution kernels (see Fig. 3.28), and the image border just described is similar to the padding we discussed in Section 3.4 and 3.5. The operations are different in morphology, but the padding and sliding operations are the same as in convolution.

In addition to the set definitions given in Section 2.6, the concept of set reflection and translation are used extensively in morphology in connection with structuring elements. The *reflection* of a set (structuring element) B about its origin, denoted by $\hat{B}$, is defined as

[†] Sets are shown as drawings of objects (e.g. squares and triangles) of arbitrary shape. A graphical image contains sets that have been embedded into a background to form a rectangular array. When we intend for a drawing to be interpreted as a *digital* image (or structuring element), we include a grid in illustrations that might otherwise be ambiguous. Objects in all drawings are shaded, and the background is shown in white. When working with actual binary images, we say that objects are *foreground* pixels. All other pixels are *background*.

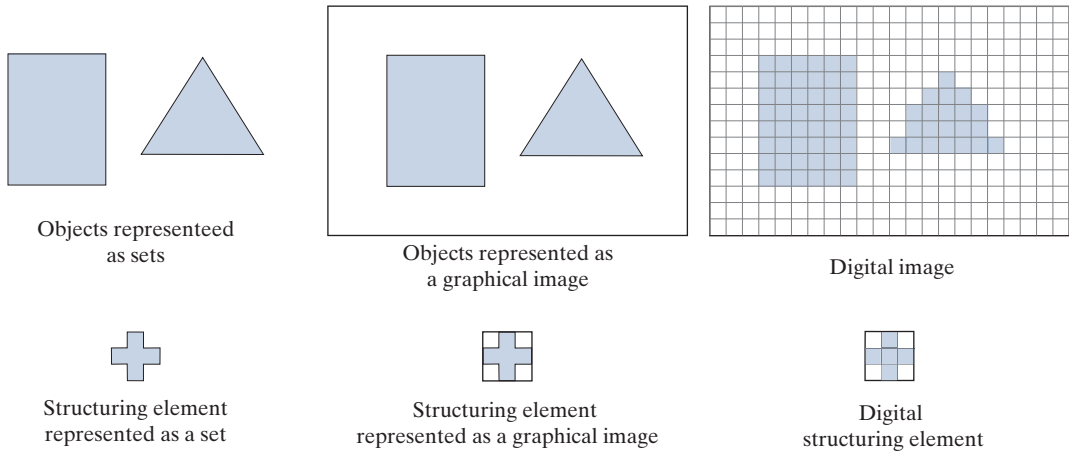


FIGURE 9.1 Top row. *Left:* Objects represented as graphical sets. *Center:* Objects embedded in a background to form a graphical image. *Right:* Object and background are digitized to form a digital image (note the grid). Second row: Example of a structuring element represented as a set, a graphical image, and finally as a digital SE.

$$\hat{B} = \{w \mid w = -b, \text{ for } b \in B\} \quad (9-1)$$

That is, if B is a set of points in 2-D, then $\hat{B}$ is the set of points in B whose (x, y) coordinates have been replaced by $(-x, -y)$. Figure 9.2 shows several examples of digital sets (structuring elements) and their reflection. The dot denotes the origin of the SE. Note that reflection consists simply of rotating an SE by 180° about its origin, and that all elements, including the background and don't care elements, are rotated.

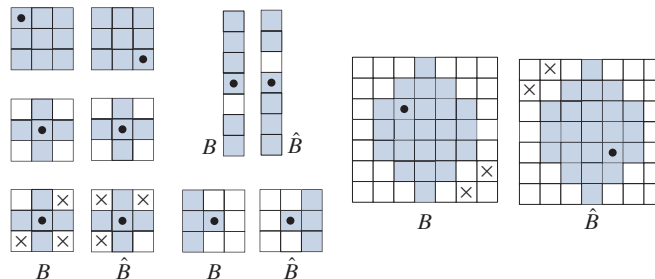
The *translation* of a set B by point $z = (z_1, z_2)$, denoted $(B)_z$, is defined as

$$(B)_z = \{c \mid c = b + z, \text{ for } b \in B\} \quad (9-2)$$

That is, if B is a set of pixels in 2-D, then $(B)_z$ is the set of pixels in B whose (x, y) coordinates have been replaced by $(x + z_1, y + z_2)$. This construct is used to translate (slide) a structuring element over an image, and each location perform a set

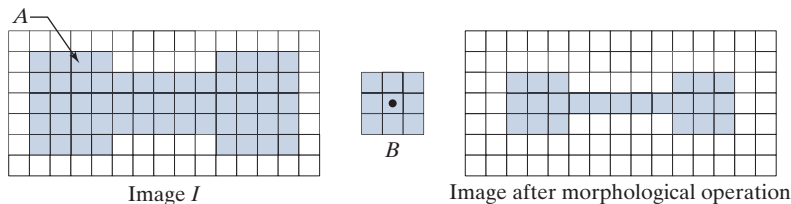
Reflection is the same operation we performed with kernels prior to spatial convolution, as explained in Section 3.4.

FIGURE 9.2 Structuring elements and their reflections about the origin (the $\times$'s are don't care elements, and the dots denote the origin). Reflection is rotation by 180° of an SE about its origin.



a b c

FIGURE 9.3
(a) A binary image containing one object (set), A . (b) A structuring element, B . (c) Image resulting from a morphological operation (see text).



operation between the structuring element and the area of the image directly under it, as we explained in Fig. 3.28 for correlation and convolution. Both reflection and translation are defined with respect to the *origin* of B .

As an introduction to how morphological operations between images and structuring elements are performed, consider Fig 9.3, which shows a simple binary image, I , consisting of an object (set) A , shown shaded, and a 3×3 SE whose elements are all 1's (foreground pixels). The background pixels (0's) are shown in white. We are interested in performing the following morphological operations: (1) form a new image, of the same size as I , consisting only of background values initially, (2) translate (slide) B over image I , and (3) at each increment of translation, if B is *completely* contained in A , mark the *location* of the origin of B as a *foreground* pixel in the new image; otherwise, leave it as a *background* point. Figure 9.3(c) is the result after the origin of B has visited every element of I . We see that, when the origin of B is on a border element of A , part of B ceases to be contained in A , thus eliminating that location of the origin of B as a possible foreground point of the new image. The net result is that the boundary of set A is *eroded*, as Fig. 9.3(e) shows. Because of the way in which we defined the operation, the maximum excursion needed for B in I is when the origin of B (which is at its center) is contained in A . With B being of size 3×3 , the narrowest background padding we needed was one pixel wide, as shown in Fig. 9.3(a). By using the smallest border needed for an operation, we keep the drawings smaller. In practice, we specify the width of padding based on the maximum dimensions of the structuring elements used, regardless of the operations being performed.

When we use terminology such as “the structuring element B is contained in set A ,” we mean *specifically* that the *foreground* elements of B overlap *only* elements of A . This becomes an important issue when B also contains background and, possibly, don't care elements. Also, we use set A to denote *all* foreground pixels of I . Those foreground elements can be a *single* object, as in Fig. 9.3, or they can represent *disjoint* subsets of foreground elements, as in the first row of Fig. 9.1. We will discuss binary images and structuring elements from Sections 9.2 through 9.7. Then, in Section 9.8, we will extend the binary ideas to grayscale images and structuring elements.

9.2 EROSION AND DILATION

We begin the discussion of morphology by studying two operations: *erosion* and *dilation*. These operations are fundamental to morphological processing. In fact, many of the morphological algorithms discussed in this chapter are based on these two primitive operations.

The reason we generally specify the padding border to be of the same dimensions as B , is that some morphological operations are defined for an entire structuring element, and cannot be interpreted with respect to the location of its origin.

EROSION

Remember, set A can represent (be the union of) multiple disjoint sets of foreground pixels (i.e., objects).

Morphological expressions are written in terms of structuring elements and a set, A , of foreground pixels, or in terms of structuring elements and an image, I , that contains A . We consider the former approach first. With A and B as sets in Z^2 , the *erosion* of A by B , denoted $A \ominus B$, is defined as

$$A \ominus B = \{z \mid (B)_z \subseteq A\} \quad (9-3)$$

where A is a set of foreground pixels, B is a structuring element, and the z 's are foreground values (1's). In words, this equation indicates that the erosion of A by B is the set of all points z such that B , translated by z , is contained in A . (Remember, displacement is defined with respect to the *origin* of B .) Equation (9-3) is the formulation that resulted in the *foreground* pixels of the image in Fig. 9.3(c).

As noted, we work with sets of foreground pixels embedded in a set of background pixels to form a complete image, I . Thus, inputs and outputs of our morphological procedures are images, not individual sets. We *could* make this fact explicit by writing Eq. (9-3) as

$$I \ominus B = \{z \mid (B)_z \subseteq A \text{ and } A \subseteq I\} \cup \{A^c \mid A^c \subseteq I\} \quad (9-4)$$

where I is a rectangular array of foreground and background pixels. The contents of the first braces say the same thing as Eq. (9-3), with the added clarification that A is a subset of (i.e., is contained in) I . The union with the operation inside the second set of braces “adds” the pixels that are not in subset A (i.e., A^c , which is the set of background pixels) to the result from the first braces, requiring also that the background pixels be part of the rectangle defined by I . In words, all this equation says is that erosion of I by B is the set of all points, z , such that B , translated by z , is contained in A . The equation also makes explicit that A is contained in I , that the result is embedded in a set of background pixels, and that the entire process is of the same size as I .

Of course, we do not use the cumbersome notation of Eq. (9-4), which we show only to emphasize an important point. Instead, we use the notation $A \ominus B$ when a morphological operation uses *only* foreground elements, and $I \ominus B$ when the operation uses foreground *and* background elements. This distinction may seem trivial, but suppose that we want to perform erosion with Eq. (9-3), using the foreground elements of the structuring element in the last column in Fig. 9.2. This structuring element also has background elements, but Eq. (9-3) assumes that B only has foreground elements. In fact, erosion is *defined* only for operations between foreground elements, so writing $I \ominus B$ would be meaningless without the “explanation” embedded in Eq. (9-4). To avoid confusion, we use A in morphological expressions when the operation involves only foreground elements, and I when the operation also involves background and/or “don’t-care” elements. We also avoid using standard morphological symbols like $\ominus$ when working with “mixed” SEs. For example, later in Eq. (9-17) we use the symbol $\otimes$ in the expression $I \otimes B = \{z \mid (B)_z \subseteq I\}$, which has the same *form* as Eq. (9-3), but instead involves an entire image and the mixed-value SE in the last column of Fig. 9.2. As you will see, using SE’s with mixed values adds considerable power to morphological operations.

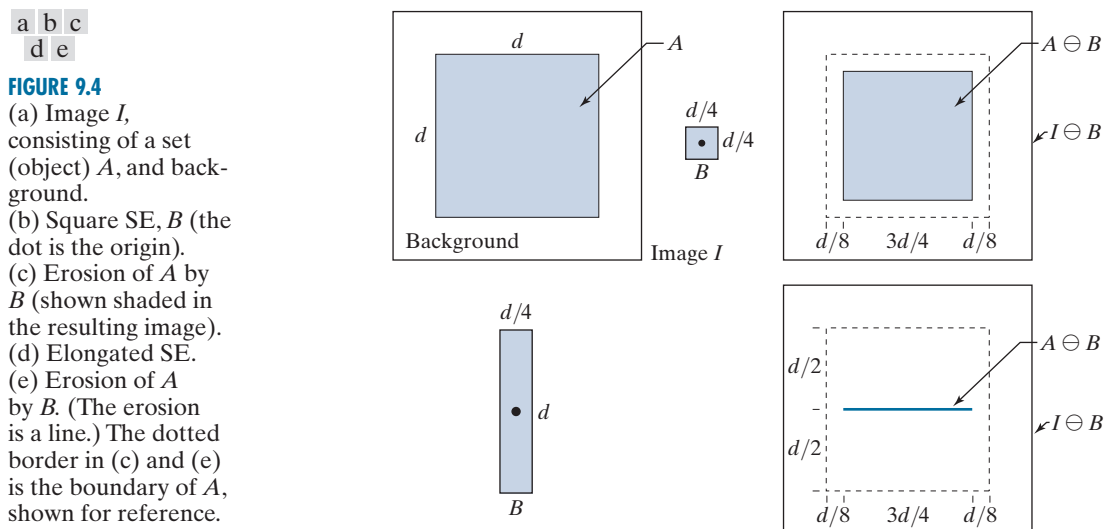
Returning to our discussion of Eq. (9-3), because the statement that B has to be contained in A is equivalent to B not sharing any common elements with the background (i.e., the set complement of A), we can express erosion equivalently as

$$A \ominus B = \{z \mid (B)_z \cap A^c = \emptyset\} \quad (9-5)$$

where, as defined in Section 2.6, $\emptyset$ is the empty set.

Figure 9.4 shows an example of erosion. The elements of set A (shaded) are the foreground pixels of image I , and, as before, the background is shown in white. The solid boundary inside the dashed boundary in Fig. 9.4(c) is the limit beyond which further displacements of the origin of B would cause some elements of the structuring element to cease being completely contained in A . Thus, the locus of points (locations of the origin of B) within (and including) this boundary constitutes the foreground elements of the erosion of A by B . We show the resulting erosion shaded in Fig. 9.4(c), and the background as white. Erosion is the *set* of values of z that satisfy Eqs. (9-3) or (9-5). The boundary of A is shown dashed in Figs. 9.4(c) and (e) as a reference; it is not part of the erosion. Figure 9.4(d) shows an elongated structuring element, and Fig. 9.4(e) shows the erosion of A by this element. Note that the original object was eroded to a line. As you can see, the result of erosion is controlled by the shape of the structuring element. In both cases, the assumption is that the image was padded to accommodate all excursions of B , and that the result was cropped to the same size as the original image, just as we did with images processed by spatial convolution in Chapter 3.

Equations (9-3) and (9-5) are not the only definitions of erosion (see Problems 9.12 and 9.13 for two additional, equivalent definitions). However, the former equations have the advantage of being more intuitive when the structuring element B is viewed as if it were a spatial kernel that slides over a set, as in convolution.



EXAMPLE 9.1: Using erosion to remove image components.

Figure 9.5(a) is a binary image depicting a simple wire-bond mask. As mentioned previously, we generally show the foreground pixels in binary images in white and the background in black. Suppose that we want to remove the lines connecting the center region to the border pads in Fig. 9.5(a). Eroding the image (i.e., eroding the *foreground* pixels of the image) with a square structuring element of size 11×11 whose components are all 1's removed most of the lines, as Fig. 9.5(b) shows. The reason that the two vertical lines in the center were thinned but not removed completely is that their width is greater than 11 pixels. Changing the SE size to 15×15 elements and eroding the original image again did remove all the connecting lines, as Fig. 9.5(c) shows. An alternate approach would have been to erode the image in Fig. 9.5(b) again, using the same 11×11 , or smaller, SE. Increasing the size of the structuring element even more would eliminate larger components. For example, the connecting lines and the border pads can be removed with a structuring element of size 45×45 elements applied to the original image, as Fig. 9.5(d) shows.

We see from this example that erosion shrinks or thins objects in a binary image. In fact, we can view erosion as a *morphological filtering* operation in which image details smaller than the structuring element are filtered (removed) from the image. In Fig. 9.5, erosion performed the function of a “line filter.” We will return to the concept of morphological filters in Sections 9.4 and 9.8.

DILATION

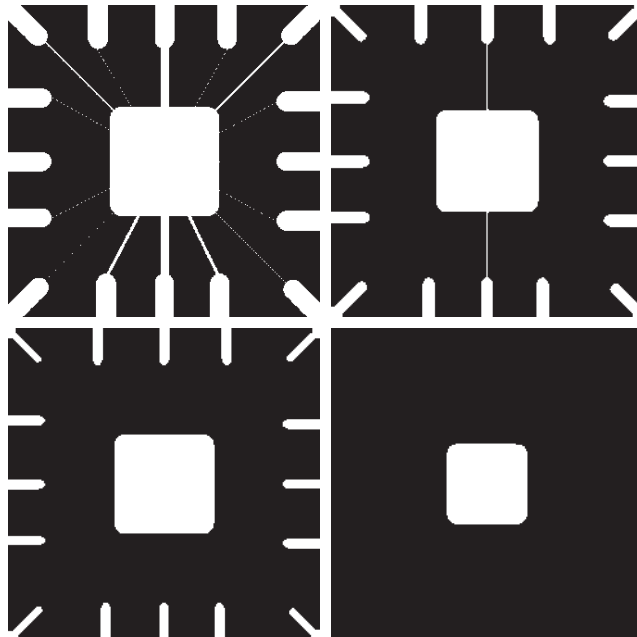
With A and B as sets in Z^2 , the dilation of A by B , denoted as $A \oplus B$, is defined as

$$A \oplus B = \left\{ z \mid (\hat{B})_z \cap A \neq \emptyset \right\} \quad (9-6)$$

a	b
c	d

FIGURE 9.5

Using erosion to remove image components. (a) A 486×486 binary image of a wire-bond mask in which foreground pixels are shown in white. (b)–(d) Image eroded using square structuring elements of sizes 11×11 , 15×15 , and 45×45 elements, respectively, all valued 1.



This equation is based on reflecting B about its origin and translating the reflection by z , as in erosion. The dilation of A by B then is the set of all displacements, z , such that the foreground elements of $\hat{B}$ overlap at least one element of A . (Remember, z is the displacement of the origin of $\hat{B}$.) Based on this interpretation, Eq. (9-6) can be written equivalently as

$$A \oplus B = \left\{ z \mid [(\hat{B})_z \cap A] \subseteq A \right\} \quad (9-7)$$

Equations (9-6) and (9-7) are not the only definitions of dilation currently in use (see Problems 9.14 and 9.15 for two different, yet equivalent, definitions). As with erosion, the preceding definitions have the advantage of being more intuitive when structuring element B is viewed as a convolution kernel. As noted earlier, the basic process of flipping (rotating) B about its origin and then successively displacing it so that it slides over set A is analogous to spatial convolution. However, keep in mind that dilation is based on set operations and therefore is a nonlinear operation, whereas convolution is a sum of products, which is a linear operation.

Unlike erosion, which is a shrinking or thinning operation, dilation “grows” or “thickens” objects in a binary image. The manner and extent of this thickening is controlled by the shape and size of the structuring element used. Figure 9.6(a) shows the same object used in Fig. 9.4 (the background area is larger to accommodate the dilation), and Fig. 9.6(b) shows a structuring element (in this case $\hat{B} = B$ because the SE is symmetric about its origin). The dashed line in Fig. 9.6(c) shows the boundary of the original object for reference, and the solid line shows the limit beyond which any further displacements of the origin of $\hat{B}$ by z would cause the intersection of $\hat{B}$ and A to be empty. Therefore, all points on and inside this boundary constitute the dilation of A by B . Figure 9.6(d) shows a structuring element designed to achieve more dilation vertically than horizontally, and Fig. 9.6(e) shows the dilation achieved with this element.

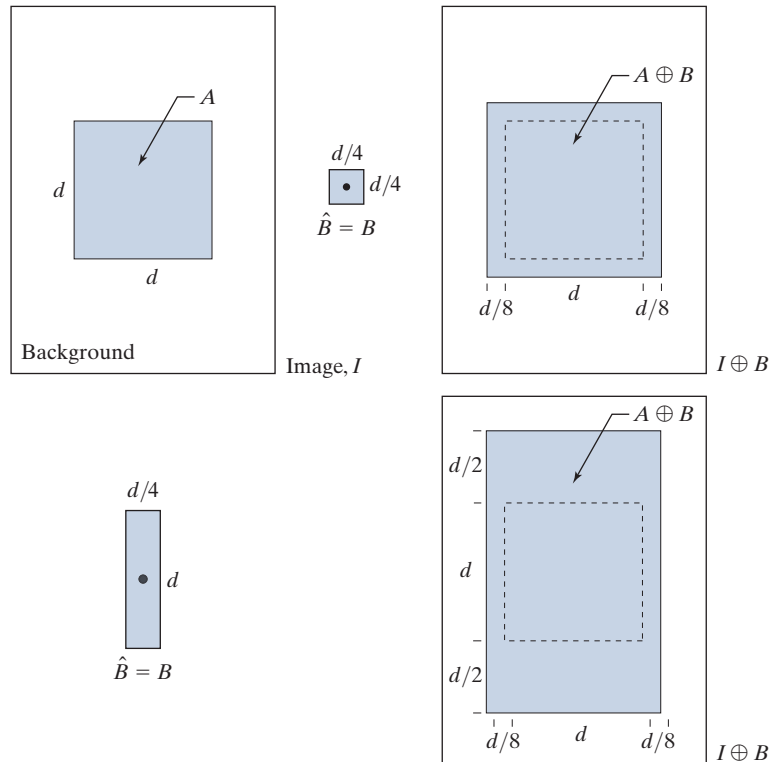
EXAMPLE 9.2: Using dilation to repair broken characters in an image.

One of the simplest applications of dilation is for bridging gaps. Figure 9.7(a) shows the same image with broken characters that we studied in Fig. 4.48 in connection with lowpass filtering. The maximum length of the breaks is known to be two pixels. Figure 9.7(b) shows a structuring element that can be used for repairing the gaps. As noted earlier, we use white (1) to denote the foreground and black (0) for the background when working with images. Figure 9.7(c) shows the result of dilating the original image with the structuring element. The gaps were bridged. One important advantage of the morphological approach over the lowpass filtering method we used to bridge the gaps in Fig. 4.48 is that the morphological method resulted directly in a binary image. Lowpass filtering, on the other hand, started with a binary image and produced a grayscale image that would require thresholding to convert it back to binary form (we will discuss thresholding in Chapter 10). Observe that set A in this application consists of numerous disjointed objects of foreground pixels.

a	b	c
d	e	

FIGURE 9.6

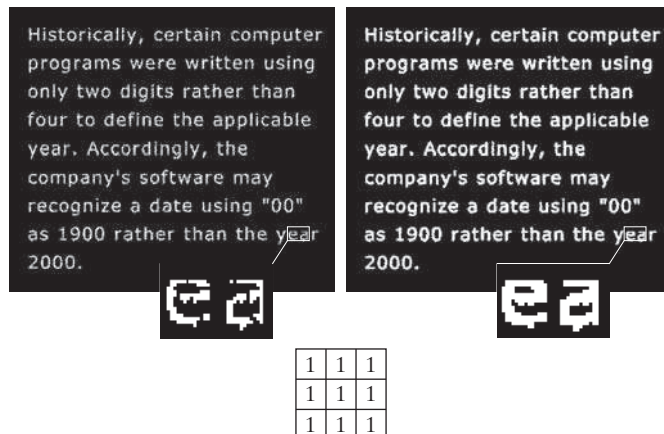
(a) Image I , composed of set (object) A and background.
 (b) Square SE (the dot is the origin).
 (c) Dilation of A by B (shown shaded).
 (d) Elongated SE.
 (e) Dilation of A by this element. The dotted line in (c) and (e) is the boundary of A , shown for reference.



a	c
b	

FIGURE 9.7

(a) Low-resolution text showing broken characters (see magnified view).
 (b) Structuring element.
 (c) Dilation of (a) by (b). Broken segments were joined.



DUALITY

Erosion and dilation are *duals* of each other with respect to set complementation and reflection. That is,

$$(A \ominus B)^c = A^c \oplus \hat{B} \quad (9-8)$$

and

$$(A \oplus B)^c = A^c \ominus \hat{B} \quad (9-9)$$

Equation (9-8) indicates that erosion of A by B is the complement of the dilation of A^c by $\hat{B}$, and vice versa. The duality property is useful when the structuring element values are symmetric with respect to its origin (as often is the case), so that $\hat{B} = B$. Then, we can obtain the erosion of A simply by dilating its background (i.e., dilating A^c) with the same structuring element and complementing the result. Similar comments apply to Eq. (9-9).

We proceed to prove formally the validity of Eq. (9-8) in order to illustrate a typical approach for establishing the validity of morphological expressions. Starting with the definition of erosion, it follows that

$$(A \ominus B)^c = \left\{ z \mid (B)_z \subseteq A \right\}^c$$

If set $(B)_z$ is contained in A , then it follows that $(B)_z \cap A^c = \emptyset$, in which case the preceding expression becomes

$$(A \ominus B)^c = \left\{ z \mid (B)_z \cap A^c = \emptyset \right\}^c$$

But the *complement* of the set of z 's that satisfy $(B)_z \cap A^c = \emptyset$ is the set of z 's such that $(B)_z \cap A^c \neq \emptyset$. Therefore,

$$\begin{aligned} (A \ominus B)^c &= \left\{ z \mid (B)_z \cap A^c \neq \emptyset \right\} \\ &= A^c \oplus \hat{B} \end{aligned}$$

where the last step follows from the definition of dilation in Eq. (9-6) and its equivalent form in Eq. (9-7). This concludes the proof. A similar line of reasoning can be used to prove Eq. (9-9) (see Problem 9.16).

9.3 OPENING AND CLOSING

As you saw in the previous section, dilation expands the components of a set and erosion shrinks it. In this section, we discuss two other important morphological operations: opening and closing. Opening generally smoothes the contour of an object, breaks narrow isthmuses, and eliminates thin protrusions. Closing also tends

to smooth sections of contours, but, as opposed to opening, it generally fuses narrow breaks and long thin gulfs, eliminates small holes, and fills gaps in the contour.

The *opening* of set A by structuring element B , denoted by $A \circ B$, is defined as

$$A \circ B = (A \ominus B) \oplus B \quad (9-10)$$

Thus, the opening A by B is the erosion of A by B , followed by a dilation of the result by B .

Similarly, the *closing* of set A by structuring element B , denoted $A \bullet B$, is defined as

$$A \bullet B = (A \oplus B) \ominus B \quad (9-11)$$

which says that the closing of A by B is simply the dilation of A by B , followed by erosion of the result by B .

Equation (9-10) has a simple geometrical interpretation: The opening of A by B is the union of all the translations of B so that B fits entirely in A . Figure 9.8(a) shows an image containing a set (object) A and Fig. 9.8(b) is a solid, circular structuring element, B . Figure 9.8(c) shows some of the translations of B such that it is contained within A , and the set shown shaded in Fig. 9.8(d) is the union of all such possible translations. Observe that, in this case, the opening is a set composed of two disjoint subsets, resulting from the fact that B could not fit in the narrow segment in the center of A . As you will see shortly, the ability to eliminate regions narrower than the structuring element is one of the key features of morphological opening.

The interpretation that the opening of A by B is the union of all the translations of B such that B fits entirely within A can be written in equation form as

$$A \circ B = \bigcup \left\{ (B)_z \mid (B)_z \subseteq A \right\} \quad (9-12)$$

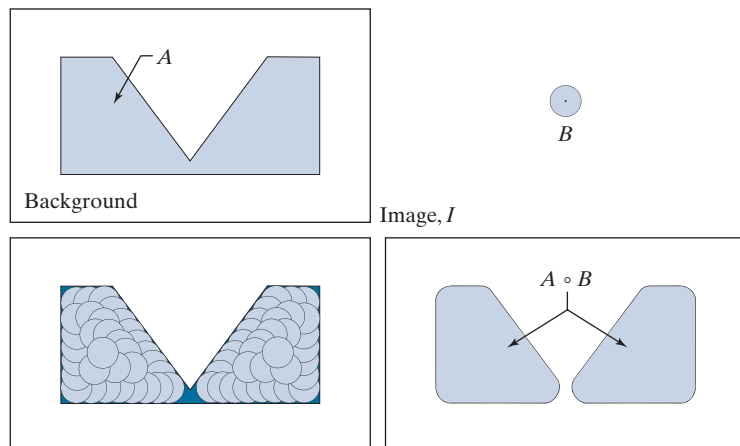
where $\bigcup$ denotes the union of the sets inside the braces.

When a circular structuring element is used for opening, the analogy is often made of the shape of the opening being determined by a “rolling ball” reaching as far as it can on the inner boundary of a set. For morphological closing the ball rolls outside, and the shape of the closing is determined by how far the ball can reach into the boundary.

a b
c d

FIGURE 9.8

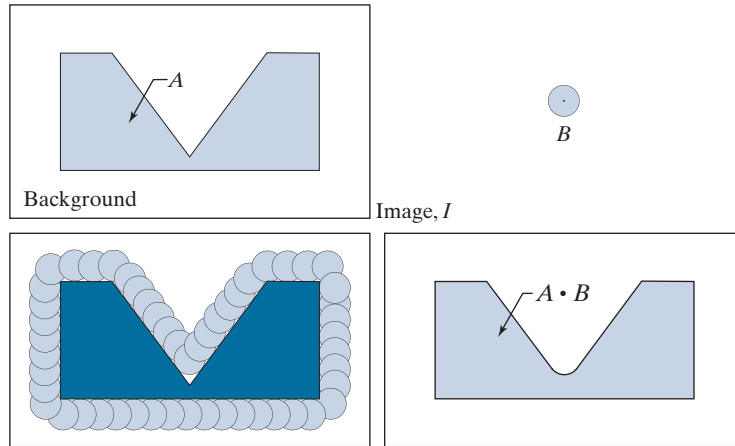
- (a) Image I , composed of set (object) A and background.
(b) Structuring element, B .
(c) Translations of B while being contained in A . (A is shown dark for clarity.)
(d) Opening of A by B .



a	b
c	d

FIGURE 9.9

(a) Image I , composed of set (object) A , and background.
 (b) Structuring element B .
 (c) Translations of B such that B does not overlap any part of A . (A is shown dark for clarity.)
 (d) Closing of A by B .



Closing has a similar geometric interpretation, except that now we translate B *outside* A . The closing is then the *complement* of the union of all translations of B that *do not* overlap A . Figure 9.9 illustrates this concept. Note that the boundary of the closing is determined by the furthest points B could reach without going inside any part of A . Based on this interpretation, we can write the closing of A by B as

$$A \bullet B = \left[\bigcup \{ (B)_z \mid (B)_z \cap A = \emptyset \} \right]^c \quad (9-13)$$

EXAMPLE 9.3: Morphological opening and closing.

Figure 9.10 shows in more detail the process and properties of opening and closing. Unlike Figs. 9.8 and 9.9, whose main objectives are overall geometrical interpretations, this figure shows the individual processes and also pays more attention to the relationship between the scale of the final results and the size of the structuring elements.

Figure 9.10(a) shows an image containing a single object (set) A , and a disk structuring element. Figure 9.10(b) shows various positions of the structuring element during erosion. This process resulted in the disjoint set in Fig. 9.10(c). Note how the bridge between the two main sections was eliminated. Its width was thin in relation to the diameter of the structuring element, which could not be completely contained in this part of the set, thus violating the definition of erosion. The same was true of the two rightmost members of the object. Protruding elements where the disk did not fit were eliminated. Figure 9.10(d) shows the process of dilating the eroded set, and Fig. 9.10(e) shows the final result of opening. Morphological opening removes regions that cannot contain the structuring element, smooths object contours, breaks thin connections, and removes thin protrusions.

Figures 9.10(f) through (i) show the results of closing A with the same structuring element. As with opening, closing also smooths the contours of objects. However, unlike opening, closing tends to join narrow breaks, fills long thin gulfs, and fills objects smaller than the structuring element. In this example, the principal result of closing was that it filled the small gulf on the left of set A .

As with erosion and dilation, opening and closing are duals of each other with respect to set complementation and reflection:

$$(A \circ B)^c = (A^c \bullet \hat{B}) \quad (9-14)$$

and

$$(A \bullet B)^c = (A^c \circ \hat{B}) \quad (9-15)$$

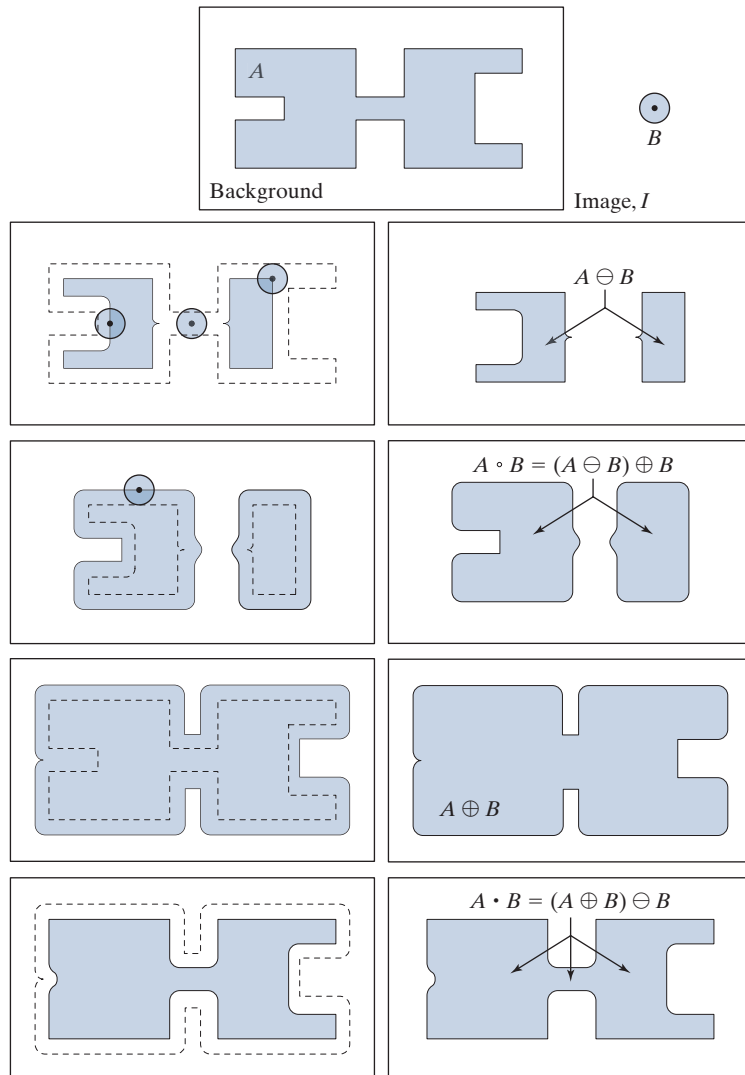
We leave the proof of these equations as an exercise (see Problem 9.20).

a
b c
d e
f g
h i

FIGURE 9.10

Morphological opening and closing.

(a) Image I , composed of a set (object) A and background; a solid, circular structuring element is shown also. (The dot is the origin.)
 (b) Structuring element in various positions.
 (c)-(i) The morphological operations used to obtain the opening and closing.



Morphological opening has the following properties:

- (a) $A \circ B$ is a subset of A .
- (b) If C is a subset of D , then $C \circ B$ is a subset of $D \circ B$.
- (c) $(A \circ B) \circ B = A \circ B$.

Similarly, closing satisfies the following properties:

- (a) A is a subset of $A \bullet B$.
- (b) If C is a subset of D , then $C \bullet B$ is a subset of $D \bullet B$.
- (c) $(A \bullet B) \bullet B = A \bullet B$.

Note from condition (c) in both cases that multiple openings or closings of a set have no effect after the operation has been applied once.

EXAMPLE 9.4: Using opening and closing for morphological filtering.

Morphological operations can be used to construct filters similar in concept to the spatial filters discussed in Chapter 3. The binary image in Fig. 9.11(a) shows a section of a fingerprint corrupted by noise. In terms of our previous notation, A is the set of all foreground (white) pixels, which includes objects of interest (the fingerprint ridges) as well as white specks of random noise. The background is black, as before. The noise manifests itself as white specks on a dark background and dark specks on the white components of the fingerprint. The objective is to eliminate the noise and its effects on the print, while distorting it as little as possible. A morphological filter consisting of an opening followed by a closing can be used to accomplish this objective.

Figure 9.11(b) shows the structuring element we used. The rest of Fig. 9.11 shows the sequence of steps in the filtering operation. Figure 9.11(c) is the result of eroding A by B . The white speckled noise in the background was eliminated almost completely in the erosion stage of opening because in this case most noise components are smaller than the structuring element. The size of the noise elements (dark spots) contained within the fingerprint actually increased in size. The reason is that these elements are inner boundaries that increase in size as objects are eroded. This enlargement is countered by performing dilation on Fig. 9.11(c). Figure 9.11(d) shows the result.

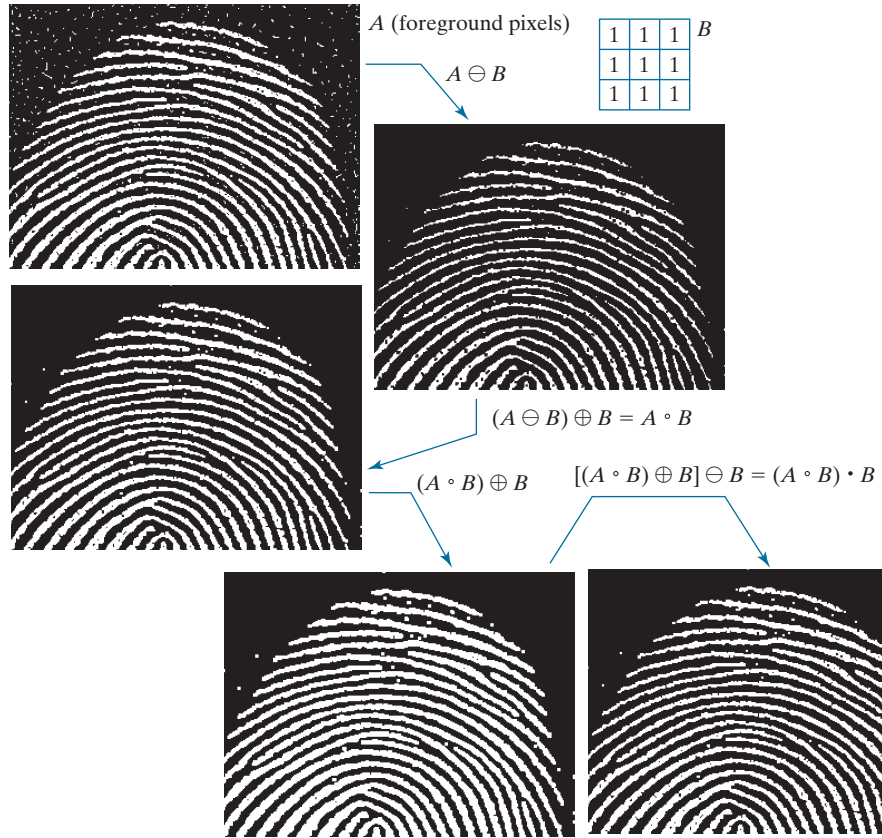
The two operations just described constitute the opening of A by B . We note in Fig. 9.11(d) that the net effect of opening was to reduce all noise components in both the background and the fingerprint itself. However, new gaps between the fingerprint ridges were created. To counter this undesirable effect, we perform a dilation on the opening, as shown in Fig. 9.11(e). Most of the breaks were restored, but the ridges were thickened, a condition that can be remedied by erosion. The result, shown in Fig. 9.11(f), is the closing of the opening of Fig. 9.11(d). This final result is remarkably clean of noise specks, but it still shows some specks of noise that appear as single pixels. These could be eliminated by methods we will discuss later in this chapter.

9.4 THE HIT-OR-MISS TRANSFORM

The morphological *hit-or-miss transform* (HMT) is a basic tool for shape detection. Let I be a binary image composed of foreground (A) and background pixels, respectively. Unlike the morphological methods discussed thus far, the HMT utilizes *two*

**FIGURE 9.11**

(a) Noisy image.
 (b) Structuring element.
 (c) Eroded image.
 (d) Dilation of the erosion (opening of A). (e) Dilation of the opening.
 (f) Closing of the opening.
 (Original image courtesy of the National Institute of Standards and Technology.)



With reference to the explanation of Eq. (9-4), we show the morphological HMT operation working directly on image I , to make it explicit that the structuring elements work on sets of foreground and background pixels simultaneously.

structuring elements: B_1 , for detecting shapes in the foreground, and B_2 , for detecting shapes in the background. The HMT of image I is defined as

$$\begin{aligned} I \circledast B_{1,2} &= \left\{ z \mid (B_1)_z \subseteq A \text{ and } (B_2)_z \subseteq A^c \right\} \\ &= (A \ominus B_1) \cap (A^c \ominus B_2) \end{aligned} \quad (9-16)$$

where the second line follows from the definition of erosion in Eq. (9-3). In words, this equation says that the morphological HMT is the set of translations, z , of structuring elements B_1 and B_2 such that, *simultaneously*, B_1 found a match in the foreground (i.e., B_1 is contained in A) and B_2 found a match in the background (i.e., B_2 is contained in A^c). The word “simultaneous” implies that z is the *same* translation of both structuring elements. The word “miss” in the HMT arises from the fact that B_2 finding a match in A^c is the same as B_2 not finding (missing) a match in A .

Figure 9.12 illustrates the concepts just introduced. Suppose that we want to find the location of the origin of object (set) D in image I . Here, A is the union of all object sets, so D is a subset of A . The need for two structuring elements capable

a	b
c	d
e	f

FIGURE 9.12

(a) Image consisting of a foreground (1's) equal to the union, A , of set of objects, and a background of 0's.

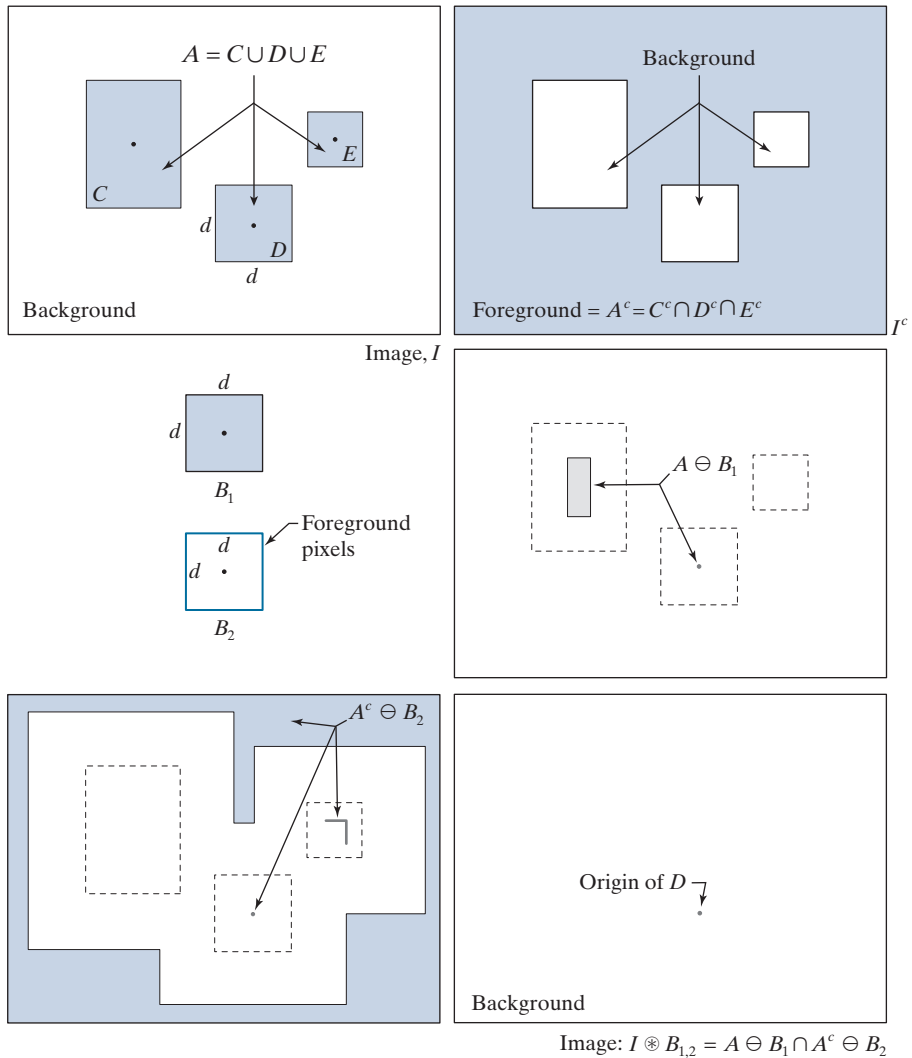
(b) Image with its foreground defined as A^c .

(c) Structuring elements designed to detect object D .

(d) Erosion of A by B_1 .

(e) Erosion of A^c by B_2 .

(f) Intersection of (d) and (e), showing the location of the origin of D , as desired. The dots indicate the origin of their respective components. Each dot is a single pixel.



of detecting properties of both the foreground and background becomes immediately obvious. All three objects are composed of foreground pixels, and one way of explaining why they appear as different shapes is because each occupies a different area of the background. In other words, the nature of a shape is determined by the geometrical arrangement of both foreground and background pixels.

Figure 9.12(a) shows that I is composed of foreground (A) and background pixels. Figure 9.12(b) is I^c , the complement of I . The foreground of I^c is defined as the set of pixels in A^c , and the background is the union of the complement of the three objects. Figure 9.12(c) shows the two structuring elements needed to detect D . Element B_1 is equal to D itself. As Fig. 9.12(d) shows, the erosion of A by B_1 contains a single point: the origin of D , as desired, but it also contains parts of object C .

Structuring element B_2 is designed to detect D in I^c . Because D is composed of background elements in I^c , and erosion works with foreground elements, B_2 has to be designed to detect the *border* of D , which is composed of foreground pixels in I^c . The SE in Fig. 9.12(c) does precisely this. It consists of a rectangle of foreground elements one pixel thick. The size of the rectangle is such that it encloses the size of D . Figure 9.12(e) shows (shaded) the erosion of the foreground of I^c by B_2 . It contains the origin of D , but is also contains parts of sets A^c and C . (The outer shaded area in Fig. 9.12(e) is larger than shown (see Problem 9.25); the result was cropped to the same size as image I for consistency.) The only elements that are common in Figs. 9.12(d) and (e) is the origin of D , so the intersection of these two sets of elements gives the location of that point, as desired. Figure 9.12(f) shows the final result.

The preceding explanation is the classic way of presenting the HMT using erosion, which is defined only for foreground pixels. A good question at this point is: Why not try to detect D directly in image I using a single structuring element, instead of going through such a laborious process? The answer is that it is possible to do so, but not in the “traditional” context of erosion the way we defined it in Eqs. (9-3) and (9-5). In order to detect D directly in image I , we would have to be able to process foreground and background pixels *simultaneously*, rather than processing just foreground pixels, as required by the definition of erosion.

To show how this can be done for the example in Fig. 9.12, we define a structuring element, B , identical to D , but having in addition a border of *background* elements with a width of one pixel. We can use a structuring element formed in such a way to restate the HMT as

$$I \circledast B = \{z \mid (B)_z \subseteq I\} \quad (9-17)$$

The *form* is the same as Eq. (9-3), but now we test to see if $(B)_z$ is a subset of *image* I , which is composed of *both* foreground and background pixels. This formulation is general, in the sense that B can be structured to detect any arrangement of pixels in image I , as Figs. 9.13 and 9.14 will illustrate.

Figure 9.13 shows graphically the same solution as Fig. 9.12(f), but using the single structuring element discussed in the previous paragraph. Figure 9.14 shows several examples based on using Eq. (9-17). The first row shows the result of using a small SE composed of both foreground (shaded) and background elements. This SE is designed to detect one-pixel holes (i.e., one background pixel surrounded by a connected border of foreground pixels) contained in image I . The SE in the second row is capable of detecting the foreground corner pixel of the top, right corner of the object in I . Using this SE in Eq. (9-17) yielded the image on the right. As you can see, the correct pixel was identified. The last row of Fig. 9.14 is more interesting, as it shows a structuring element composed of foreground, background, and “don’t care” elements which, as mentioned earlier, we denote by $\times$ ’s. You can think of the value of a don’t care element as always matching its corresponding pixel in an image. In this example, when the SE is centered on the top, right corner pixel, the don’t care elements in the top of the SE can be considered to be background, and the don’t care elements on the bottom row as foreground, producing a correct

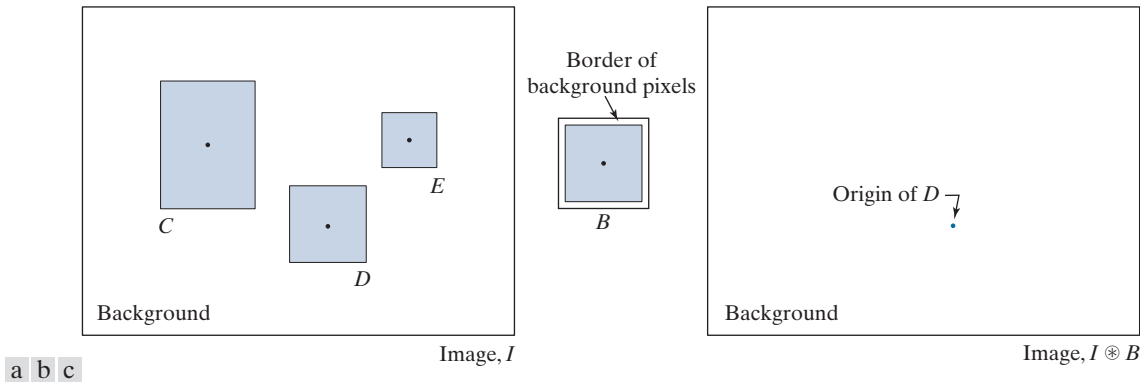


FIGURE 9.13 Same solution as in Fig. 9.12, but using Eq. (9-17) with a single structuring element.

match. When the SE is centered on the bottom, right corner pixel, the role of the don't care elements is reversed, again resulting in a correct match. The other border pixels between the two corners were similarly detected by considering all don't care elements as foreground. Thus, using don't care elements increases the flexibility of structuring elements to perform multiple roles.

9.5 SOME BASIC MORPHOLOGICAL ALGORITHMS

With the preceding discussion as a foundation, we are now ready to consider some practical uses of morphology. When dealing with binary images, one of the principal applications of morphology is in extracting image components that are useful in the

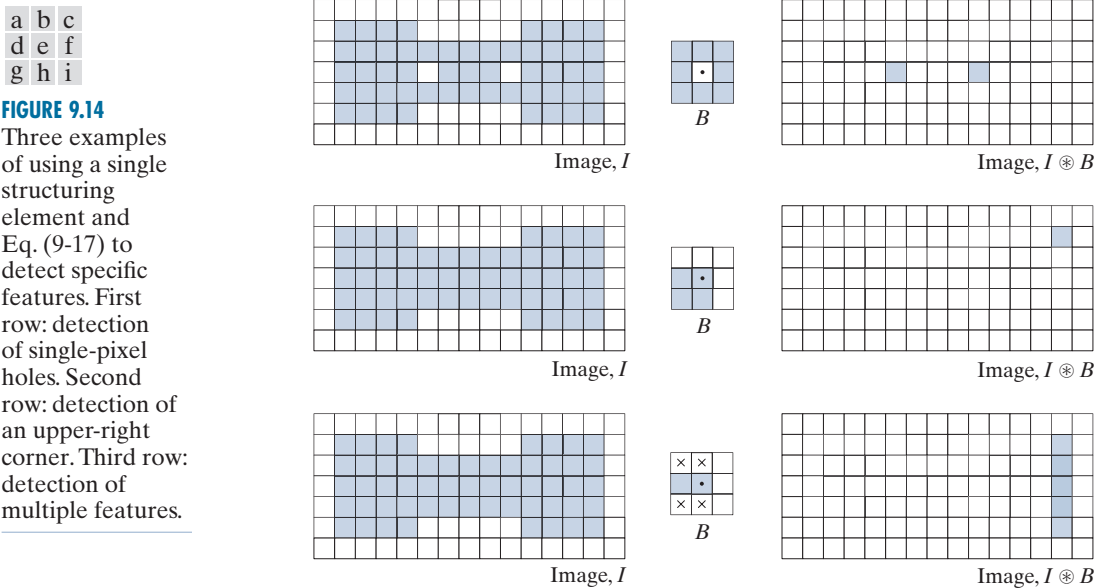


FIGURE 9.14 Three examples of using a single structuring element and Eq. (9-17) to detect specific features. First row: detection of single-pixel holes. Second row: detection of an upper-right corner. Third row: detection of multiple features.

representation and description of shape. In particular, we consider morphological algorithms for extracting boundaries, connected components, the convex hull, and the skeleton of a region. We also develop several methods (for region filling, thinning, thickening, and pruning) that are used frequently for pre- or post-processing. We make extensive use in this section of “mini-images,” designed to clarify the mechanics of each morphological method as we introduce it. These binary images are shown graphically with foreground (1’s) shaded and background (0’s) in white, as before.

BOUNDARY EXTRACTION

The boundary of a set A of foreground pixels, denoted by $\beta(A)$, can be obtained by first eroding A by a suitable structuring element B , and then performing the set difference between A and its erosion. That is,

$$\beta(A) = A - (A \ominus B) \quad (9-18)$$

Figure 9.15 illustrates the mechanics of boundary extraction. It shows a simple binary object, a structuring element B , and the result of using Eq. (9-18). The structuring element in Fig. 9.15(b) is among the most frequently used, but it is not unique. For example, using a 5×5 structuring element of 1’s would result in a boundary between 2 and 3 pixels thick. It is understood that the image in Fig. 9.15(a) was padded with a border of background elements, and that the results were cropped back to the original size after the morphological operations were completed.

EXAMPLE 9.5: Boundary extraction.

Figure 9.16 further illustrates the use of Eq. (9-18) using a 3×3 structuring element of 1’s. As before when working with images, we show foreground pixels (1’s) in white and background pixels (0’s) in black. The elements of the SE, which are 1’s, also are treated as white. Because of the size of the structuring element used, the boundary in Fig. 9.16(b) is one pixel thick.

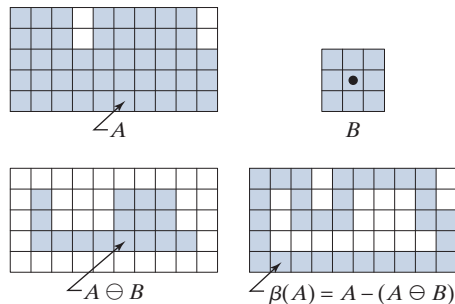
HOLE FILLING

As mentioned in the discussion of Fig. 9.14, a *hole* may be defined as a background region surrounded by a connected border of foreground pixels. In this section, we develop an algorithm based on set dilation, complementation, and intersection for

a b
c d

FIGURE 9.15

- (a) Set, A , of foreground pixels.
(b) Structuring element.
(c) A eroded by B .
(d) Boundary of A .



a b

FIGURE 9.16

(a) A binary image.
 (b) Result of using Eq. (9-18) with the structuring element in Fig. 9.15(b).



filling holes in an image. Let A denote a set whose elements are 8-connected boundaries, with each boundary enclosing a background region (i.e., a hole). Given a point in each hole, the objective is to fill all the holes with foreground elements (1's).

We begin by forming an array, X_0 , of 0's (the same size as I , the image containing A), except at locations in X_0 that correspond to pixels that are known to be holes, which we set to 1. Then, the following procedure fills all the holes with 1's:

$$X_k = (X_{k-1} \oplus B) \cap I^c \quad k = 1, 2, 3, \dots \quad (9-19)$$

where B is the symmetric structuring element in Fig. 9.17(c). The algorithm terminates at iteration step k if $X_k = X_{k-1}$. Then, X_k contains all the filled holes. The set union of X_k and I contains all the filled holes and their boundaries.

The dilation in Eq. (9-19) would fill the entire area if left unchecked, but the intersection at each step with I^c limits the result to inside the region of interest. This is our first example of how a morphological process can be *conditioned* to meet a desired property. In the current application, the process is appropriately called *conditional dilation*. The rest of Fig. 9.17 illustrates further the mechanics of Eq. (9-19). This example only has one hole, but the concept applies to any finite number of holes, assuming that a point inside each hole is given (we remove this requirement in Section 9.6).

EXAMPLE 9.6: Morphological hole filling.

Figure 9.18(a) shows an image of white circles with black holes. An image such as this might result from thresholding into two levels a scene containing polished spheres (e.g., ball bearings). The dark circular areas inside the spheres would result from reflections. The objective is to eliminate the reflections by filling the holes in the image. Figure 9.18(b) shows the result of filling all the spheres. Because it must be known whether black points are background points or sphere inner points (i.e., holes), fully automating this procedure requires that additional “intelligence” be built into the algorithm. We will give a fully automatic approach in Section 9.6 based on morphological reconstruction (see Problem 9.36 also).

Remember, the dilation of image X by B is the dilation of the foreground elements of X by B .

a	b	c
d	e	f
g	h	i

FIGURE 9.17

Hole filling.

(a) Set A (shown shaded) contained in image I .

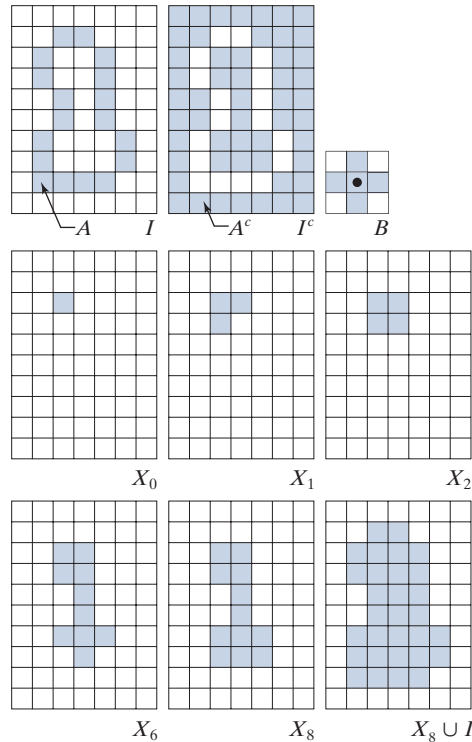
(b) Complement of I .

(c) Structuring element B . Only the foreground elements are used in computations

(d) Initial point inside hole, set to 1.

(e)–(h) Various steps of Eq. (9-19).

(i) Final result [union of (a) and (h)].



EXTRACTION OF CONNECTED COMPONENTS

Connectivity and connected components are discussed in Section 2.5.

Being able to extract connected components from a binary image is central to many automated image analysis applications. Let A be a set of foreground pixels consisting of one or more connected components, and form an image X_0 (of the same size as I , the image containing A) whose elements are 0's (background values), except at each location known to correspond to a point in each connected component in A ,

a b

FIGURE 9.18

(a) Binary image. The white dots inside the regions (shown enlarged for clarity) are the starting points for the hole-filling algorithm.

(b) Result of filling all holes.

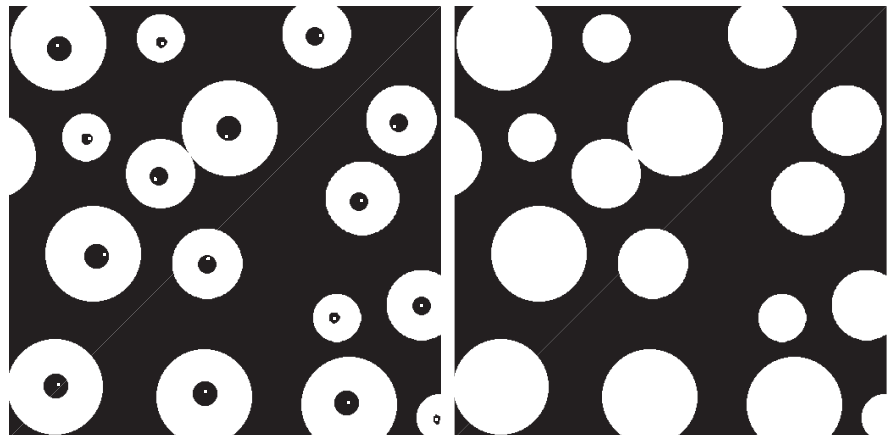
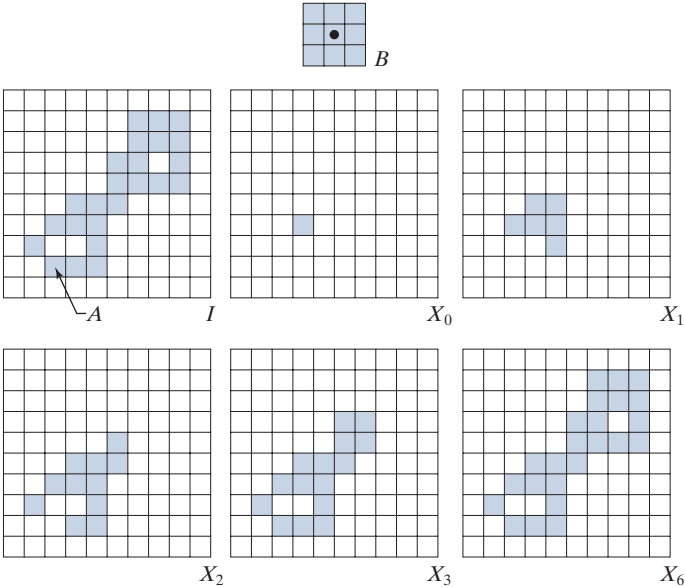




FIGURE 9.19
(a) Structuring element.
(b) Image containing a set with one connected component.
(c) Initial array containing a 1 in the region of the connected component.
(d)–(g) Various steps in the iteration of Eq. (9-20)



which we set to 1 (foreground value). The objective is to start with X_0 and find all the connected components in I . The following iterative procedure accomplishes this:

$$X_k = (X_{k-1} \oplus B) \cap I \qquad k = 1, 2, 3, \dots \tag{9-20}$$

where B is the SE in Fig. 9.19(a). The procedure terminates when $X_k = X_{k-1}$, with X_k containing all the connected components of foreground pixels in the image. Both Eqs. (9-19) and (9-20) use conditional dilation to limit the growth of set dilation, but Eq. (9-20) uses I instead of I^c . This is because here we are looking for foreground points, while the objective of (9-19) is to find background points. Figure 9.19 illustrates the mechanics of Eq. (9-20), with convergence being achieved for $k = 6$. Note that the shape of the structuring element used is based on 8-connectivity between pixels. As in the hole-filling algorithm, Eq. (9-20) is applicable to any finite number of connected components contained in I , assuming that a point is known in each. See Problem 9.37 for a completely automated procedure that removes this requirement.

EXAMPLE 9.7: Using connected components to detect foreign objects in packaged food.

Connected components are used frequently for automated inspection. Figure 9.20(a) shows an X-ray image of a chicken breast that contains bone fragments. It is important to be able to detect such foreign objects in processed foods before shipping. In this application, the density of the bones is such that their nominal intensity values are significantly different from the background. This makes extraction of the bones from the background a simple matter by using a single threshold (thresholding was introduced in Section 3.1 and we will discuss in more detail in Section 10.3). The result is the binary image in Fig. 9.20(b). The most significant feature in this figure is the fact that the points that remain after thresholding are clustered into objects (bones), rather than being scattered. We can make sure that only objects of

a
b
c d

FIGURE 9.20

(a) X-ray image of a chicken fillet with bone fragments.

(b) Thresholded image (shown as the negative for clarity).

(c) Image eroded with a 5×5 SE of 1's.

(d) Number of pixels in the connected components of (c).

(Image (a) courtesy of NTB Elektronische Geraete GmbH, Diepholz, Germany, www.ntbxray.com.)



Connected component	No. of pixels in connected comp
01	11
02	9
03	9
04	39
05	133
06	1
07	1
08	743
09	7
10	11
11	11
12	9
13	9
14	674
15	85

“significant” size are contained in the binary image by eroding its foreground. In this example, we define as significant any object that remains after erosion with a 5×5 SE of 1's. Figure 9.20(c) shows the result of erosion. The next step is to analyze the size of the objects that remain. We label (identify) these objects by extracting the connected components in the image. The table in Fig. 9.20(d) lists the results of the extraction. There are 15 connected components, with four of them being dominant in size. This is enough evidence to conclude that significant, undesirable objects are contained in the original image. If needed, further characterization (such as shape) is possible using the techniques discussed in Chapter 11.

CONVEX HULL

A set, S , of points in the Euclidean plane is said to be *convex* if and only if a straight line segment joining any two points in S lies entirely within S . The *convex hull*, H , of S is the smallest convex set containing S . The *convex deficiency* of S is defined as the set difference $H - S$. Unlike the Euclidean plane, the digital image plane (see Fig. 2.19) only allows points at discrete coordinates. Thus, the sets with which we work are *digital sets*. The same concepts of convexity are applicable to digital sets, but the definition of a convex digital set is slightly different. A *digital set*, A , is said to be *convex* if and only if its Euclidean convex hull only contains digital points

belonging to A . A simple way to visualize if a digital set of foreground points is convex is to join its boundary points by straight (continuous) Euclidean line segments. If only foreground points are contained within the set formed by the line segments, then the set is convex; otherwise it is not. The definitions of convex hull and convex deficiency given above for S , extend directly to digital sets. The following morphological algorithm can be used to obtain an approximation of the convex hull of a set A of foreground pixels, embedded in a binary image, I .

Let B^i , $i = 1, 2, 3, 4$, denote the four structuring elements in Fig. 9.21(a). The procedure consists of implementing the morphological equation

$$X_k^i = (X_{k-1}^i \circledast B^i) \cup X_{k-1}^i \quad i = 1, 2, 3, 4 \quad \text{and} \quad k = 1, 2, 3, \dots \quad (9-21)$$

with $X_0^i = I$. When the procedure converges using the i th structuring element (i.e., when $X_k^i = X_{k-1}^i$), we let $D^i = X_k^i$. Then, the convex hull of A is the union of the four results:

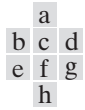
$$C(A) = \bigcup_{i=1}^4 D^i \quad (9-22)$$

Thus, the method consists of iteratively applying the hit-or-miss transform to I with B^1 until convergence, then letting $D^1 = X_k^1$, where k is the step at which convergence occurred. The procedure is repeated with B^2 (applied to I) until no further changes occur, and so on. The union of the four resulting D^i constitutes the convex hull of A . The algorithm is initialized with $k = 0$ and $X_0^i = I$ every time that i (i.e., the structuring element) changes.

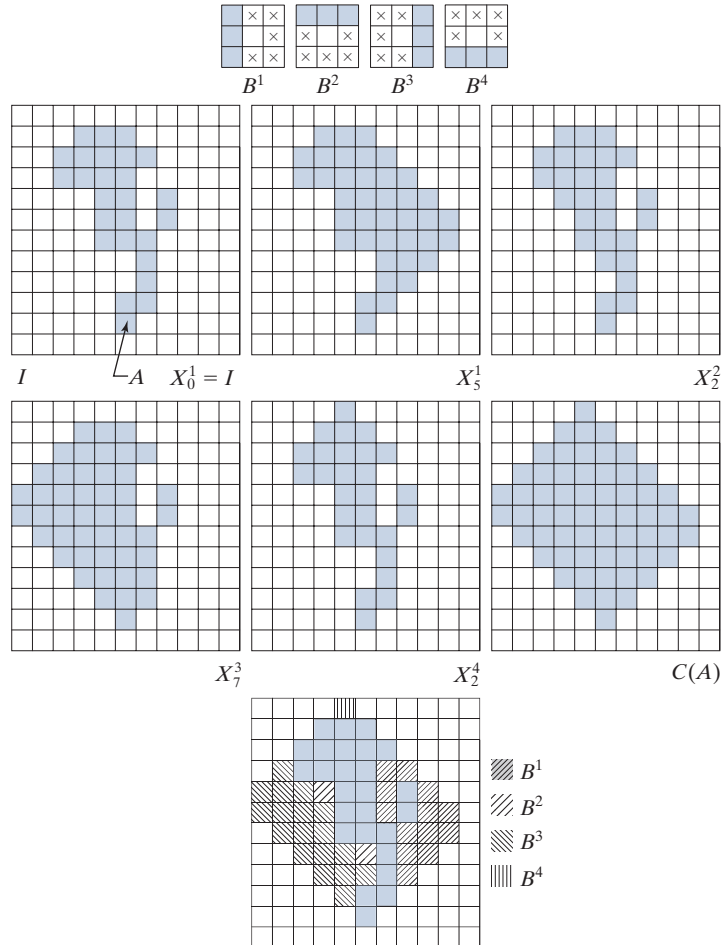
Figure 9.21 illustrates the use of Eqs. (9-21) and (9-22). Figure 9.21(a) shows the structuring elements used to extract the convex hull. The origin of each element is at its center. As before, the $\times$ entries indicate “don’t care” elements. Recall that the HMT is said to have found a match of structuring element B^i in a 3×3 region of I , if all the elements of B^i find corresponding matches in that region. As noted earlier, when computing a match, a “don’t care” element can be interpreted as always matching the value of its corresponding element in the image. Note in Fig. 9.21(a) that B^i is a clockwise rotation of B^{i-1} by 90° .

Figure 9.21(b) shows a set A for which the convex hull is sought. As before, the set is embedded in an array of background elements to form an image, I . Starting with $X_0^1 = I$ resulted in the set in Fig. 9.21(c) after five iterations of Eq. (9-21). Then, letting $X_0^2 = I$ and again using Eq. (9-21) resulted in the set in Fig. 9.21(d) (convergence was achieved in only two steps in this case). The next two results were obtained in the same manner. Finally, forming the union of the sets in Figs. 9.21(c), (d), (e), and (f) resulted in the convex hull in Fig. 9.21(g). The contribution of each structuring element is highlighted in the composite set shown in Fig. 9.21(h).

One obvious shortcoming of the procedure just discussed is that the convex hull can grow beyond the minimum dimensions required to guarantee convexity, thus violating the definition of the convex hull. This, in fact, is what happened in this case. One simple approach to reduce this growth is to place limits so that it does not extend beyond the vertical and horizontal dimensions of set A . Imposing this

**FIGURE 9.21**

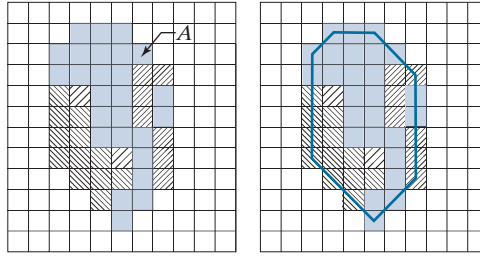
- (a) Structuring elements.
 (b) Set A .
 (c)–(f) Results of convergence with the structuring elements shown in (a).
 (g) Convex hull.
 (h) Convex hull showing the contribution of each structuring element.



a b

FIGURE 9.22

(a) Result of limiting growth of the convex hull algorithm.
 (b) Straight lines connecting the boundary points show that the new set is convex also.



THINNING

Thinning of a set A of foreground pixels by a structuring element B , denoted $A \otimes B$, can be defined in terms of the hit-or-miss transform:

$$\begin{aligned} A \otimes B &= A - (A \circledast B) \\ &= A \cap (A \circledast B)^c \end{aligned} \quad (9-23)$$

where the second line follows from the definition of set difference given in Eq. (2-40). A more useful expression for thinning A symmetrically is based on a *sequence* of structuring elements:

$$\{B\} = \{B^1, B^2, B^3, \dots, B^n\} \quad (9-24)$$

Using this concept, we now define thinning by a sequence of structuring elements as

$$A \otimes \{B\} = \left(\left(\dots \left((A \otimes B^1) \otimes B^2 \right) \dots \right) \otimes B^n \right) \quad (9-25)$$

The process is to thin A by one pass with B^1 , then thin the result with one pass of B^2 , and so on, until A is thinned with one pass of B^n . The entire process is repeated until no further changes occur after one complete pass through all structuring elements. Each individual thinning pass is performed using Eq. (9-23).

Figure 9.23(a) shows a set of structuring elements used routinely for thinning (note that B^i is equal to B^{i-1} rotated clockwise by 45°), and Fig. 9.23(b) shows a set A to be thinned, using the procedure just discussed. Figure 9.23(c) shows the result of thinning A with one pass of B^1 to obtain A_1 . Figure 9.23(d) is the result of thinning A_1 with B^2 , and Figs. 9.23(e) through (k) show the results of passes with the remaining structuring elements (there were no changes from A_7 to A_8 or from A_9 to A_{11} .) Convergence was achieved after the second pass of B^6 . Figure 9.23(l) shows the thinned result. Finally, Fig. 9.23(m) shows the thinned set converted to m -connectivity (see Section 2.5 and Problem 9.29) to eliminate multiple paths.

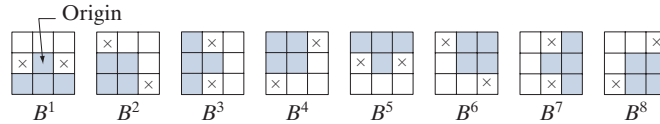
As before, we assume that the image containing A was padded to accommodate all excursions of B , and that the result was cropped. We show only A for simplicity.

THICKENING

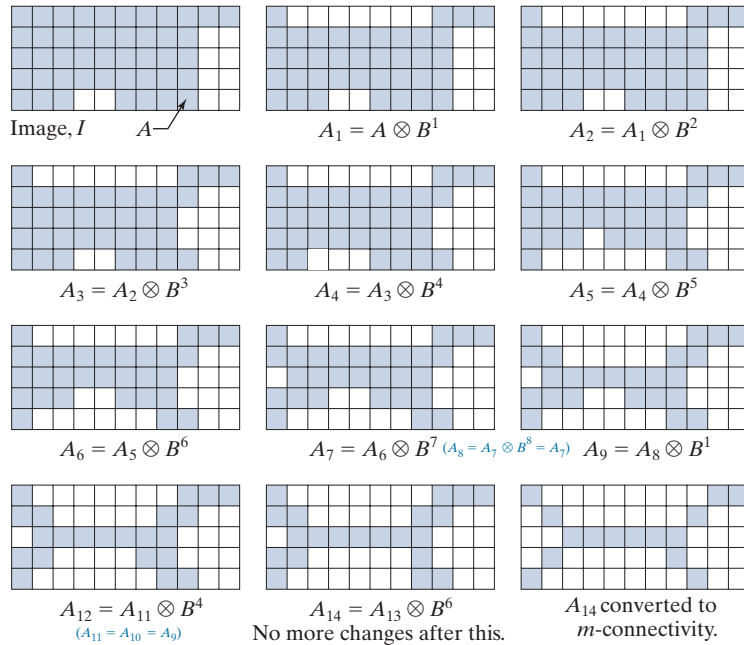
Thickening is the morphological dual of thinning and is defined by the expression

$$A \odot B = A \cup (A \circledast B) \quad (9-26)$$

a
b c d
e f g
h i j
k l m

**FIGURE 9.23**

- (a) Structuring elements.
 (b) Set A .
 (c) Result of thinning A with B^1 (shaded).
 (d) Result of thinning A_1 with B_2 .
 (e)–(i) Results of thinning with the next six SEs. (There was no change between A_7 and A_8 .)
 (j)–(k) Result of using the first four elements again.
 (l) Result after convergence.
 (m) Result converted to m -connectivity.



where B is a structuring element suitable for thickening. As in thinning, thickening can be defined as a sequential operation:

$$A \odot \{B\} = \left(\left(\left(\left(A \odot B^1 \right) \odot B^2 \right) \dots \right) \odot B^n \right) \quad (9-27)$$

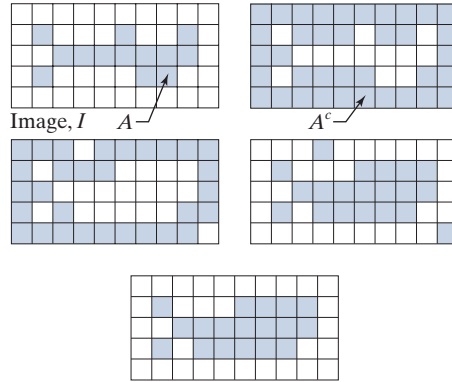
The structuring elements used for thickening have the same form as those shown in Fig. 9.23(a), but with all 1's and 0's interchanged. However, a separate algorithm for thickening is seldom used in practice. Instead, the usual procedure is to thin the background of the set in question, then complement the result. In other words, to thicken a set A we form A^c , thin A^c , and then complement the thinned set to obtain the thickening of A . Figure 9.24 illustrates this procedure. As before, we show only set A and image I , and not the padded version of I .

Depending on the structure of A , this procedure can result in disconnected points, as Fig. 9.24(d) shows. Hence thickening by this method usually is followed by post-processing to remove disconnected points. Note from Fig. 9.24(c) that the thinned background forms a boundary for the thickening process. This useful feature is not present in the direct implementation of thickening using Eq. (9-27), and it is one of the principal reasons for using background thinning to accomplish thickening.

a	b
c	d
e	

FIGURE 9.24

- (a) Set A .
 (b) Complement of A .
 (c) Result of thinning the complement.
 (d) Thickened set obtained by complementing (c).
 (e) Final result, with no disconnected points.



SKELETONS

We will discuss skeletons in more detail in Section 11.2.

As Fig. 9.25 shows, the notion of a *skeleton* $S(A)$ of a set A is intuitively simple. We deduce from this figure that

- (a) If z is a point of $S(A)$, and $(D)_z$ is the largest disk centered at z and contained in A , one cannot find a larger disk (not necessarily centered at z) containing $(D)_z$ and simultaneously included in A . A disk $(D)_z$ satisfying these conditions is called a *maximum disk*.
 (b) If $(D)_z$ is a maximum disk, it touches the boundary of A at two or more different places.

The skeleton of A can be expressed in terms of erosions and openings. That is, it can be shown (Serra [1982]) that

$$S(A) = \bigcup_{k=0}^K S_k(A) \quad (9-28)$$

with

$$S_k(A) = (A \ominus kB) - (A \ominus kB) \circ B \quad (9-29)$$

where B is a structuring element, and $(A \ominus kB)$ indicates k successive erosions starting with A ; that is, A is first eroded by B , the result is eroded by B , and so on:

$$(A \ominus kB) = (((\dots((A \ominus B) \ominus B) \ominus \dots) \ominus B) \quad (9-30)$$

k times. K in Eq. (9-28) is the last iterative step before A erodes to an empty set. In other words,

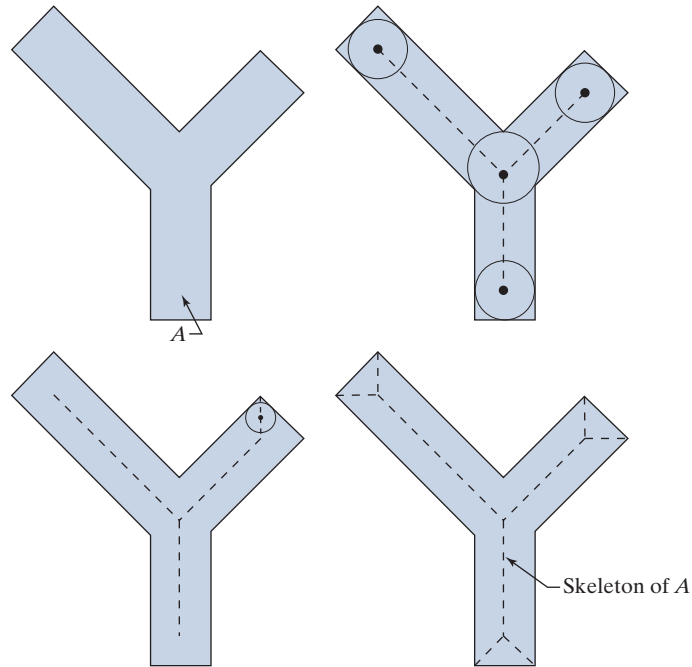
$$K = \max \{k \mid (A \ominus kB) \neq \emptyset\} \quad (9-31)$$

The formulation in Eqs. (9-28) and (9-29) indicate that $S(A)$ can be obtained as the union of the skeleton subsets $S_k(A)$, $k = 0, 1, 2, \dots, K$.

a	b
c	d

FIGURE 9.25

(a) Set A .
 (b) Various positions of maximum disks whose centers partially define the skeleton of A .
 (c) Another maximum disk, whose center defines a different segment of the skeleton of A .
 (d) Complete skeleton (dashed).



It can be shown (Serra [1982]) that A can be reconstructed from these subsets:

$$A = \bigcup_{k=0}^K (S_k(A) \oplus kB) \quad (9-32)$$

where $(S_k(A) \oplus kB)$ denotes k successive dilations, starting with $S_k(A)$; that is,

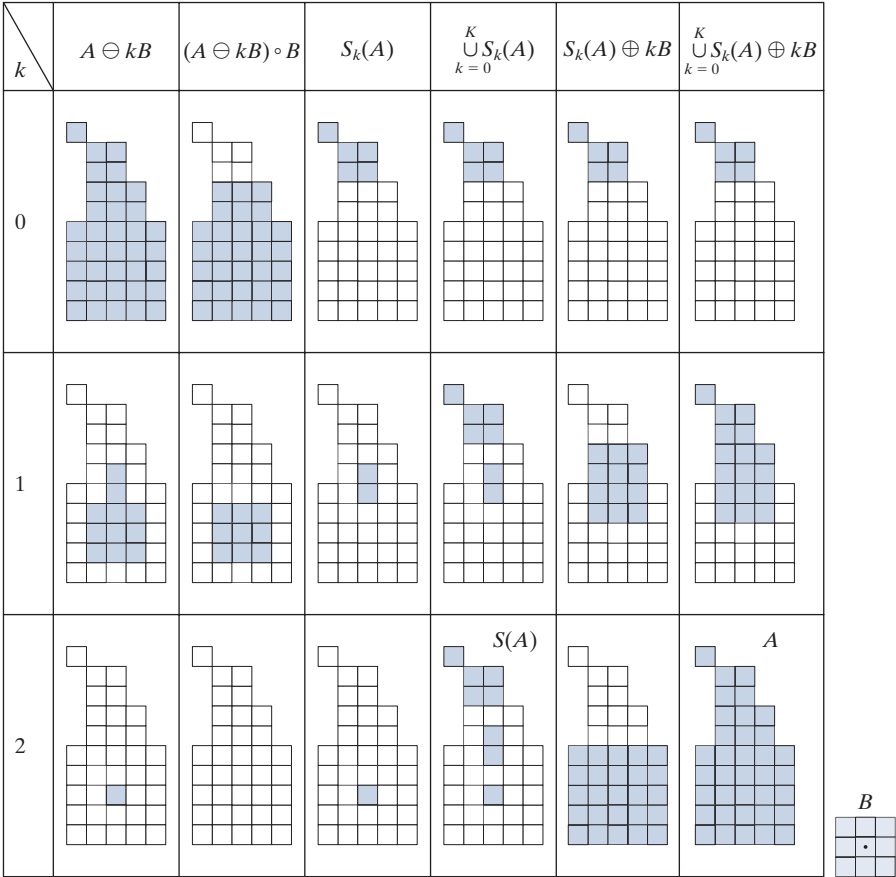
$$(S_k(A) \oplus kB) = ((\dots((S_k(A) \oplus B) \oplus B) \oplus \dots) \oplus B) \quad (9-33)$$

EXAMPLE 9.8: Computing the skeleton of a simple set.

Figure 9.26 illustrates the concepts just discussed. The first column shows the original set (at the top) and two erosions by the structuring element B shown in the figure. Note that one more erosion would yield the empty set, so $K = 2$ in this case. The second column shows the opening by B of the sets in the first column. These results are easily explained by the fitting characterization of the opening operation discussed in connection with Fig. 9.8. The third column contains the set differences between the first and second columns. Thus, the three entries in the third column are $S_0(A)$, $S_1(A)$, and $S_2(A)$, respectively.

The fourth column contains two partial skeletons, and the final result at the bottom of the column. The final skeleton not only is thicker than it needs to be but, more important, it is not connected. This result is not unexpected, as nothing in the preceding formulation of the morphological skeleton guarantees connectivity. Morphology produces an elegant formulation in terms of erosions and openings of the given set. However, heuristic formulations (see Section 11.2) are needed if, as is usually the case, the skeleton must be maximally thin, connected, and minimally eroded.

FIGURE 9.26
Implementation of Eqs. (9-28) through (9-33). The original set is at the top left, and its morphological skeleton is at the bottom of the fourth column. The reconstructed set is at the bottom of the sixth column.



The entries in the fifth and sixth columns deal with reconstructing the original set from its skeleton subsets. The fifth column are the dilations of $S_k(A)$; that is, $S_0(A)$, $S_1(k) \oplus B$, and $S_2(A) \oplus 2B = (S_2(A) \oplus B) \oplus B$. Finally, the last column shows reconstruction of set A which, according to Eq. (9-32), is the union of the dilated skeleton subsets shown in the fifth column.

PRUNING

Pruning methods are an essential complement to thinning and skeletonizing algorithms, because these procedures tend to leave *spurs* (“parasitic” components) that need to be “cleaned up” by postprocessing. We begin the discussion with a pruning problem, then develop a solution based on the material introduced in the preceding sections. Thus, we take this opportunity to illustrate how to solve a problem by combining several of the morphological techniques discussed up to this point.

A common approach in the automated recognition of hand-printed characters is to analyze the shape of the skeleton of a character. These skeletons often contain spurs, caused during erosion by noise and non-uniformities in the character strokes.

We may define an *end point* as the center point of a 3×3 region that satisfies any of the arrangements in Fig. 9.27(b).

In this section we develop a morphological technique for handling this problem, starting with the assumption that the length of a parasitic component does not exceed a specified number of pixels.

Figure 9.27(a) shows the skeleton of a hand-printed letter “a.” The spur on the leftmost part of the character exemplifies what we are interested in removing. The solution is based on suppressing a spur branch by successively eliminating its end point. Of course, this also shortens (or eliminates) other branches in the character but, in the absence of other structural information, the assumption in this example is that any branch with three or less pixels is to be eliminated. Thinning of a set A , with a sequence of structuring elements designed to detect only end points, achieves the desired result. That is, let

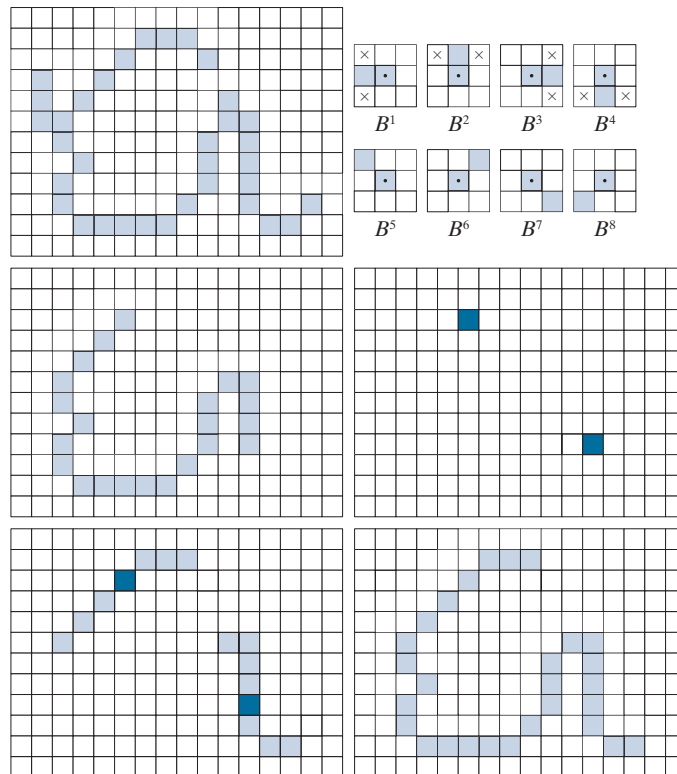
$$X_1 = A \otimes \{B\} \quad (9-34)$$

where $\{B\}$ denotes the structuring element sequence in Fig. 9.27(b) [see Eq. (9-24) regarding structuring-element sequences]. The sequence of structuring elements consists of two different structures, each of which is rotated 90° for a total of eight elements. The $\times$ in Fig. 9.27(b) signifies a “don’t care” condition, as defined earlier. (Note that each SE is a detector for an end point in a particular orientation.)

a b
c d
e f

FIGURE 9.27

- (a) Set A of foreground pixels (shaded).
- (b) SEs used for deleting end points.
- (c) Result of three cycles of thinning.
- (d) End points of (c).
- (e) Dilation of end points conditioned on (a).
- (f) Pruned image.



Applying Eq. (9-34) to A three times yielded the set X_1 in Fig. 9.27(c). The next step is to “restore” the character to its original form, but with the parasitic branches removed. This requires that we first form a set X_2 containing all end points in X_1 [Fig. 9.27(e)]:

$$X_2 = \bigcup_{k=1}^8 (X_1 \otimes B^k) \quad (9-35)$$

where the B^k are the end-point detectors in Fig. 9.27(b). The next step is dilation of the end points. Typically, the number of dilations is less than the number of end-point removals to reduce the probability of “growing” back some of the spurs. In this case, we know by inspection that no new spurs are created, so we dilate the end points three times using A as a delimiter. This is the same number of thinning passes:

$$X_3 = (X_2 \oplus H) \cap A \quad (9-36)$$

where H is a 3×3 structuring element of 1's, and the intersection with A is applied after each step. As in the case of region filling, this type of conditional dilation prevents the creation of 1-valued elements outside the region of interest, as illustrated by the result in Fig. 9.27(e). Finally, the union of X_1 and X_3 ,

$$X_4 = X_1 \cup X_3 \quad (9-37)$$

yields the desired result in Fig. 9.27(f).

In more complex scenarios, using Eq. (9-36) sometimes picks up the “tips” of some branches. This can occur when the end points of these branches are near the skeleton. Although Eq. (9-36) may eliminate them, they can be picked up again during dilation because they are valid points in A . However, unless entire parasitic elements are picked up again (a rare case if these elements are short with respect to valid strokes), detecting and eliminating the reconstructed elements is easy because they are disconnected regions.

A natural thought at this juncture is that there must be easier ways to solve this problem. For example, we could just keep track of all deleted points and simply reconnect the appropriate points to all end points left after application of Eq. (9-34). This argument is valid, but the advantage of the formulation just presented is that we used existing morphological constructs to solve the problem. When a set of such tools is available, the advantage is that no new algorithms have to be written. We simply combine the necessary morphological functions into a sequence of operations.

Sometimes you will encounter end point detectors based on a single structuring element, similar to the first SE in Fig. 9.27(b), but having “don't care” conditions along the entire first column instead having a foreground element separating the corner $\times$'s. This is incorrect. For example, the former element would identify the point located in the eighth row, fourth column of Fig. 9.27(a) as an end point, thus eliminating it and breaking the connectivity of that part of the stroke.

9.6 MORPHOLOGICAL RECONSTRUCTION

The morphological concepts discussed thus far involve a single image and one or more structuring elements. In this section, we discuss a powerful morphological transformation called *morphological reconstruction* that involves two images and a structuring element. One image, the *marker*, which we denote by F , contains the starting points for reconstruction. The other image, the *mask*, denoted by G , constrains (conditions) the reconstruction. The structuring element is used to define connectivity.[†] For 2-D applications, connectivity typically is defined as 8-connectivity, which is implied by a structuring element of size 3×3 whose elements are all 1's.

See Section 2.5 regarding connectivity.

GEODESIC DILATION AND EROSION

Central to morphological reconstruction are the concepts of geodesic dilation and geodesic erosion. Let F denote the marker image and G the mask image. We assume in this discussion that both are binary images and that $F \subseteq G$. The *geodesic dilation of size 1* of the marker image with respect to the mask, denoted by $D_G^{(1)}(F)$, is defined as

$$D_G^{(1)}(F) = (F \oplus B) \cap G \quad (9-38)$$

where, as usual, $\cap$ denotes the set intersection (here $\cap$ may be interpreted as a logical AND because we are dealing with binary quantities). The *geodesic dilation of size n* of F with respect to G is defined as

$$D_G^{(n)}(F) = D_G^{(1)}(D_G^{(n-1)}(F)) \quad (9-39)$$

where $n \geq 1$ is an integer, and $D_G^{(0)}(F) = F$. In this recursive expression, the set intersection indicated in Eq. (9-38) is performed at each step.[‡] Note that the intersection operation guarantees that mask G will limit the growth (dilation) of marker F . Figure 9.28 shows a simple example of a geodesic dilation of size 1. The steps in the figure are a direct implementation of Eq. (9-38). Note that the marker F consists of just one point from the object in G . The idea is to grow (dilate) this point successively, masking of the result at each step by G . Continuing with this process would yield a result whose shape is influenced by the structure of G . In this simple case, the reconstruction would eventually result in an image identical to G (see Fig. 9.30).

The *geodesic erosion of size 1* of marker F with respect to mask G is defined as

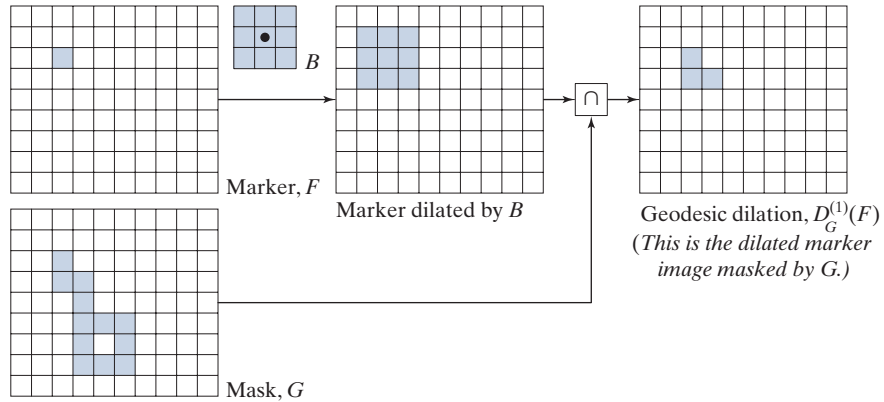
$$E_G^{(1)}(F) = (F \ominus B) \cup G \quad (9-40)$$

[†] In much of the literature on morphological reconstruction, the structuring element is tacitly assumed to be isotropic and typically is called an *elementary isotropic structuring element*. In the context of this chapter, an example of such an SE is a 3×3 array of 1's with the origin at the center.

[‡] Although it is more intuitive to develop morphological reconstruction methods using recursive formulations (as we do here), their practical implementation typically is based on more computationally efficient algorithms (see, for example, Vincent [1993] and Soille [2003]).

FIGURE 9.28

Illustration of a geodesic dilation of size 1. Note that the marker image contains a point from the object in G . If continued, subsequent dilations and maskings would eventually result in the object contained in G .



where $\cup$ denotes set union (or logical OR operation). The geodesic erosion of size n of F with respect to G is defined as

$$E_G^{(n)}(F) = E_G^{(1)}(E_G^{(n-1)}(F)) \quad (9-41)$$

where $n \geq 1$ is an integer and $E_G^{(0)}(F) = F$. The set union in Eq. (9-40) is performed at each step, and guarantees that geodesic erosion of an image remains greater than or equal to its mask image. As you might have expected from the forms in Eqs. (9-38) and (9-40), geodesic dilation and erosion are duals with respect to set complementation (see Problem 9.41). Figure 9.29 shows an example of a geodesic erosion of size 1. The steps in the figure are a direct implementation of Eq. (9-40).

Geodesic dilation and erosion converge after a finite number of iterative steps, because propagation or shrinking of the marker image is constrained by the mask.

MORPHOLOGICAL RECONSTRUCTION BY DILATION AND BY EROSION

Based on the preceding concepts, *morphological reconstruction by dilation* of a marker image F with respect to a mask image G , denoted $R_G^D(F)$, is defined as the geodesic dilation of F with respect to G , iterated until stability is achieved; that is,

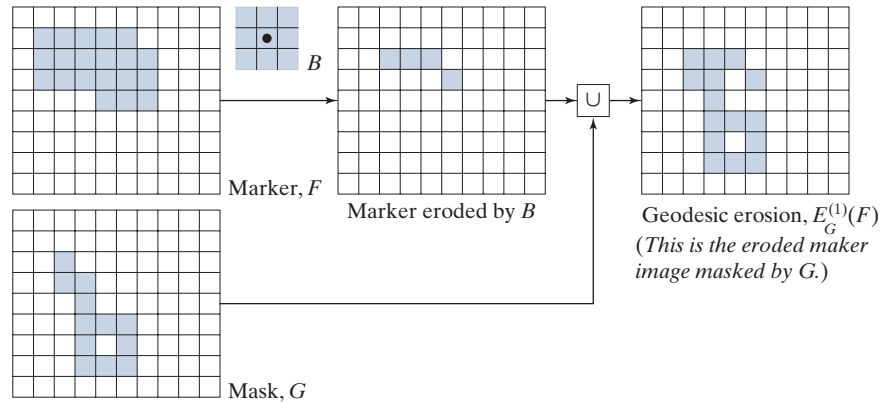
$$R_G^D(F) = D_G^{(k)}(F) \quad (9-42)$$

with k such that $D_G^{(k)}(F) = D_G^{(k+1)}(F)$.

Figure 9.30 illustrates reconstruction by dilation. Figure 9.30(a) continues the process begun in Fig. 9.28. The next step in reconstruction after obtaining $D_G^{(1)}(F)$ is to dilate this result, then AND it with mask G to yield $D_G^{(2)}(F)$, as Fig. 9.30(b) shows. Dilation of $D_G^{(1)}(F)$ and masking with G then yields $D_G^{(3)}(F)$, and so on. This procedure is repeated until stability is reached. Carrying out this example one more step would give $D_G^{(5)}(F) = D_G^{(6)}(F)$, so the image, morphologically reconstructed by dilation, is given by $R_G^D(F) = D_G^{(5)}(F)$, as indicated in Eq. (9-42). The reconstructed image is identical to the mask, as expected.

FIGURE 9.29

Illustration of a geodesic erosion of size 1.



In a similar manner, the *morphological reconstruction by erosion* of a marker image F with respect to a mask image G , denoted $R_G^E(F)$, is defined as the geodesic erosion of F with respect to G , iterated until stability; that is,

$$R_G^E(F) = E_G^{(k)}(F) \quad (9-43)$$

with k such that $E_G^{(k)}(F) = E_G^{(k+1)}(F)$. As an exercise, generate a figure similar to Fig. 9.30 for morphological reconstruction by erosion. Reconstruction by dilation and erosion are duals with respect to set complementation (see Problem 9.42).

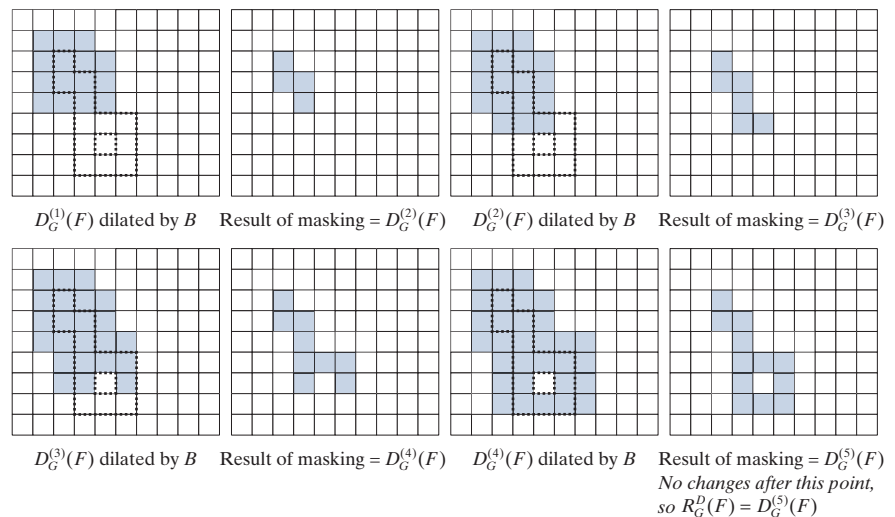
SAMPLE APPLICATIONS

Morphological reconstruction has a broad spectrum of practical applications, each determined by the selection of the marker and mask images, by the structuring

a	b	c	d
e	f	g	h

FIGURE 9.30

Illustration of morphological reconstruction by dilation. Sets $D_G^{(1)}(F)$, G , B and F are from Fig. 9.28. The mask (G) is shown dotted for reference.



elements, and by combinations of the morphological operations defined in the preceding discussion. The following examples illustrate the usefulness of these concepts.

Opening by Reconstruction

In morphological opening, erosion removes small objects and then dilation attempts to restore the shape of the objects that remain. The accuracy of this restoration depends on the similarity of the shapes and the structuring element(s) used. Opening by reconstruction restores exactly the shapes of the objects that remain after erosion. The *opening by reconstruction* of size n of an image F is defined as the reconstruction by dilation of the erosion of size n of F with respect to F ; that is,

$$O_R^{(n)}(F) = R_F^D(F \ominus nB) \quad (9-44)$$

where $F \ominus nB$ indicates n erosions by B starting with F , as defined in Eq. (9-30). Note that F itself is used as the mask. By comparing this equation with Eq. (9-42), we see that Eq. (9-44) indicates that the opening by reconstruction uses an eroded version of F as the marker in reconstruction by dilation.

As you will see in Fig. 9.31, Eq. (9-44) can lead to some interesting results. Typically, the structuring element, B , used in Eq. (9-44) is designed to extract some feature of interest, based on erosion. However, as mentioned at the beginning of this section, the structuring element used in reconstruction (i.e., in the dilation that is performed to obtain R_F^D) is designed to define connectivity and, for 2-D, that structuring element typically is a 3×3 array of 1's. It is important that you do not confuse this SE with the structuring element, B , used for erosion in Eq. (9-44). Finally, we point out that this equation is most commonly used with $n = 1$.

Figure 9.31 shows an example of opening by reconstruction. We are interested in extracting from Fig. 9.31(a) the characters that contain long, vertical strokes. This objective determines the nature of B in Eq. (9-44). The average height of the tall characters in the figure is 51 pixels. By eroding the image with a thin structuring element of size 51×1 , we should be able to isolate these characters. Figure 9.31(b) shows one erosion [$n = 1$ in Eq. (9-44)] of Fig. 9.31(a) with the structuring element just mentioned. As you can see, the locations of the tall characters were extracted successively. For the purpose of comparison, we computed the opening (remember this is erosion followed by the dilation) of the image using the same structuring element. Figure 9.31(c) shows the result. As noted earlier, simply dilating an eroded image does not always restore the original. Finally, Fig. 9.31(d) is the reconstruction by dilation of the original image using that image as the mask and the eroded image as the marker. The dilation in the reconstruction was done using a 3×3 SE of 1's, for the reason mentioned earlier. Because we only performed one erosion, the steps just followed constitute the opening by reconstruction (of size 1) of F [i.e., $O_R^{(1)}(F)$] given in Eq. (9-44). As the figure shows, characters containing long vertical strokes were restored accurately from the eroded image (i.e., the marker); all other characters were removed.

A expression similar to Eq. (9-44) can be written for *closing by reconstruction* (see Table 9.1 and Problem 9.44). The difference is that the marker used for closing by reconstruction is the dilation of F and, instead of R_F^D , we use R_F^E . As you saw,

A expression similar to this equation can be written for closing by reconstruction (see Table 9.1 and Problem 9.44).

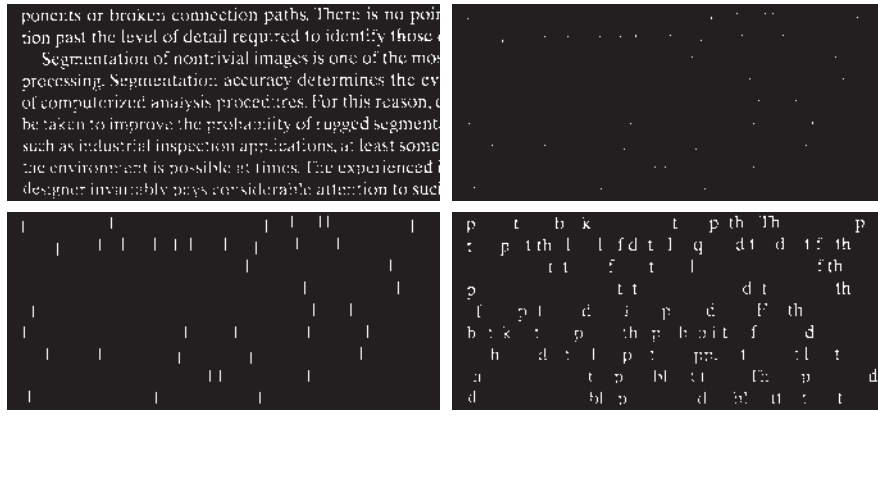


FIGURE 9.31 (a) Text image of size 918×2018 pixels. The approximate average height of the tall characters is 51 pixels. (b) Erosion of (a) with a structuring element of size 51×1 elements (all 1's). (c) Opening of (a) with the same structuring element, shown for comparison. (d) Result of opening by reconstruction.

opening by reconstruction works with images in which the background is black (0) and the foreground is white (1). Closing by reconstruction works with the opposite scenario. For example, if we were working with the complement of Fig. 9.31(a), the background would be white and the foreground black. To solve the same problem of extracting the tall characters, we would use opening by reconstruction. All the other images in Fig. 9.31 would be identical, except that they would be black on white. The structuring element used would be the *same* in both cases, so the operations of closing by reconstruction would be performed on background pixels.

Automatic Algorithm for Filling Holes

In Section 9.5, we developed an algorithm for filling holes based on knowing a starting point in each hole. Here, we develop a fully automated procedure based on morphological reconstruction. Let $I(x, y)$ denote a binary image, and suppose that we form a marker image F that is 0 everywhere, except at the image border, where it is set to $1 - I$, that is,

$$F(x, y) = \begin{cases} 1 - I(x, y) & \text{if } (x, y) \text{ is on the border of } I \\ 0 & \text{otherwise} \end{cases} \quad (9-45)$$

Then,

$$H = [R_{F^c}^D(F)]^c \quad (9-46)$$

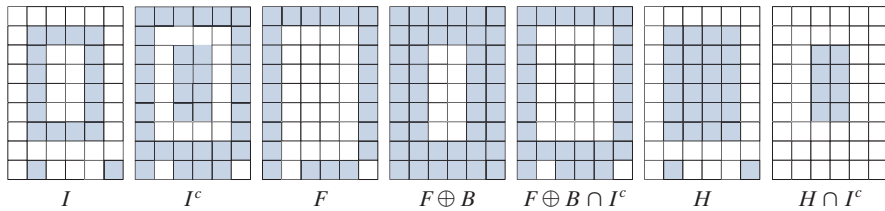
is a binary image equal to I with all holes filled.

To see how Eqs. (9-45) and (9-46) cause holes in an image to be filled, consider Figs. 9.32(a) and (b), which show an image, I , containing one hole, and the image

a b c d e f g

FIGURE 9.32

Hole filling using morphological reconstruction.



complement, respectively. The complement of I sets all foreground (1-valued) pixels to background (0-valued) pixels, and vice versa. By definition, a hole is surrounded by foreground pixels. Therefore, this operation builds a “wall” of 0’s around the hole. Because I^c is used as an AND mask, what we are doing is protecting all foreground pixels from changing during iteration. Figure 9.32(c) is array F , formed according to Eq. (9-45), and Fig. 9.32(d), using a 3×3 SE of 1’s. The marker F has a border of 1’s (except at locations where I is 1), so the dilation of the marker points starts at the border and proceeds inward. Figure 9.32(e) shows the geodesic dilation of F using I^c as the mask. We see that all locations in this result that correspond to foreground pixels of I are now 0, and that this is true for the hole pixels as well. Another iteration will yield the same result which, when complemented as required by Eq. (9-46), gives the result in Fig. 9.32(f). The hole is now filled and the rest of image I was unchanged. The operation $H \cap I^c$ yields an image containing 1-valued pixels in the locations corresponding to the holes in I and 0’s elsewhere, as Fig. 9.32(g) shows.

Figure 9.33 shows a more practical example. Figure 9.33(b) shows the complement of the text image in Fig. 9.33(a), and Fig. 9.33(c) is the marker image, F , generated using Eq. (9-45). This image is all black with a white (1’s) border, except at locations corresponding to 1’s in the border of the original image (the border values are not easily discernible by eye at the magnification shown, and also because the page is nearly white). Finally, Fig. 9.33(d) shows the image with all the holes filled.

Border Clearing

Extracting objects from an image for subsequent shape analysis is a fundamental task in automated image processing. An algorithm for detecting objects that touch (i.e., are connected to) the border is a useful tool because (1) it can be used to screen images so that only complete objects remain for further processing, or (2) it can be used as a signal that partial objects are present in the field of view. As a final illustration of the concepts introduced in this section, we develop a border-clearing procedure based on morphological reconstruction. In this application, we use the original image as the mask and the following marker image:

$$F(x, y) = \begin{cases} I(x, y) & \text{if } (x, y) \text{ is on the border of } I \\ 0 & \text{otherwise} \end{cases} \quad (9-47)$$

The border-clearing algorithm first computes the morphological reconstruction $R_I^D(F)$ (which extracts the objects touching the border), and then computes the following difference:

a b
c d

FIGURE 9.33

(a) Text image of size 918×2018 pixels.

(b) Complement of (a) for use as a mask image.

(c) Marker image.

(d) Result of hole-filling using Eqs. (9-45) and (9-46).

ponents or broken connection paths. There is no position past the level of detail required to identify those. Segmentation of nontrivial images is one of the most processing. Segmentation accuracy determines the evolution of computerized analysis procedures. For this reason, one must be taken to improve the probability of rugged segment such as industrial inspection applications, at least some the environment is possible at times. The experienced designer invariably pays considerable attention to such



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$$X = I - R_I^D(F) \quad (9-48)$$

to obtain an image, X , with no objects touching the border.

As an example, consider the original text image from Fig. 9.31(a) again. Figure 9.34(a) shows the reconstruction $R_I^D(F)$ obtained using a 3×3 structuring element of 1's. The objects touching the border of the original image are visible in the right side of Fig. 9.34(a). Figure 9.34(b) shows image X , computed using Eq. (9-48). If the task at hand were automated character recognition, having an image in which no characters touch the border is most useful because the problem of having to recognize partial characters (a difficult task at best) is avoided.

9.7 SUMMARY OF MORPHOLOGICAL OPERATIONS ON BINARY IMAGES

Figure 9.35 summarizes the types of structuring elements used in the various binary morphological methods discussed thus far. The shaded elements are foreground values (typically denoted by 1's in numerical arrays), the elements in white are background values (typically denoted by 0's), and the x's are "don't care" elements. Table 9.1 summarizes the binary morphological results developed in the preceding sections. The Roman numerals in the third column of Table 9.1 refer to the structuring elements in Fig. 9.35.

a b

FIGURE 9.34

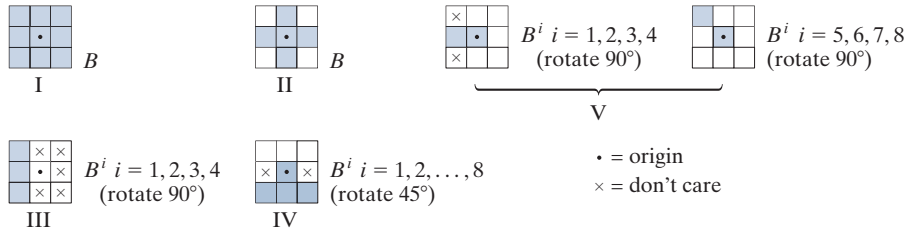
(a) Reconstruction by dilation of marker image. (b) Image with no objects touching the border. The original image is Fig. 9.31(a).



ponents or broken connection paths. There is no position past the level of detail required to identify those. Segmentation of nontrivial images is one of the most processing. Segmentation accuracy determines the evolution of computerized analysis procedures. For this reason, one must be taken to improve the probability of rugged segment such as industrial inspection applications, at least some the environment is possible at times. The experienced designer invariably pays considerable attention to such

FIGURE 9.35

Five basic types of structuring elements used for binary morphology.



9.8 GRAYSCALE MORPHOLOGY

In this section, we extend to grayscale images the basic operations of dilation, erosion, opening, and closing. We then use these operations to develop several basic grayscale morphological algorithms. Throughout the discussion that follows, we deal with digital functions of the form $f(x, y)$ and $b(x, y)$, where $f(x, y)$ is a grayscale image and $b(x, y)$ is a structuring element. The assumption is that these functions are discrete in the sense defined in Section 2.4. That is, if Z denotes the set of real integers, then the coordinates (x, y) are integers from the Cartesian product Z^2 , and $f(x, y)$ and $b(x, y)$ are functions that assign an intensity value (a real number from the set of real numbers, R) to each distinct pair of coordinates (x, y) . If the intensity levels are integers also, then Z replaces R .

Structuring elements in grayscale morphology perform the same basic functions as their binary counterparts: They are used as “probes” to examine a given image for specific properties. Structuring elements in grayscale morphology belong to one of two categories: *nonflat* and *flat*. Figure 9.36 shows an example of each. Figure 9.36(a) is a hemispherical grayscale SE shown as an image, and Fig. 9.36(c) is a horizontal intensity profile through its center. Figure 9.34(b) shows a flat structuring element in the shape of a disk, and Fig. 9.36(d) is its corresponding intensity profile. (The shape of this profile explains the origin of the word “flat.”) The elements in Fig. 9.36 are shown as continuous quantities for clarity; their computer implementation is based on digital approximations. Because of a number of difficulties discussed later in this section, grayscale nonflat SEs are not used frequently in practice. Finally, we mention that, as in the binary case, the origin of grayscale structuring elements must be clearly identified. Unless mentioned otherwise, all the examples in this section are based on symmetrical, flat structuring elements of unit height whose origins are at the center. The reflection of an SE in grayscale morphology is as defined in Section 9.1; we denote it in the following discussion by $\hat{b}(x, y) = b(-x, -y)$.

GRAYSCALE EROSION AND DILATION

The *grayscale erosion* of f by a flat structuring element b at location (x, y) is defined as the *minimum* value of the image in the region coincident with $b(x, y)$ when the origin of b is at (x, y) . In equation form, the erosion at (x, y) of an image f by a structuring element b is given as

$$[f \ominus b](x, y) = \min_{(s, t) \in b} \{f(x + s, y + t)\} \quad (9-49)$$

TABLE 9.1

Summary of binary morphological operations and their properties. A is a set of foreground pixels contained in binary image I , and B is a structuring element. I is a binary image (containing A), with 1's corresponding to the elements of A and 0's elsewhere. The Roman numerals refer to the structuring elements in Fig. 9.35.

Operation	Equation	Comments
Translation	$(B)_z = \{c \mid c = b + z, \text{ for } b \in B\}$	Translates the origin of B to point z .
Reflection	$\hat{B} = \{w \mid w = -b, \text{ for } b \in B\}$	Reflects B about its origin.
Complement	$A^c = \{w \mid w \notin A\}$	Set of points not in A .
Difference	$A - B = \{w \mid w \in A, w \notin B\}$ $= A \cap B^c$	Set of points in A , but not in B .
Erosion	$A \ominus B = \{z \mid (B)_z \subseteq A\}$	Erodes the boundary of A . (I)
Dilation	$A \oplus B = \{z \mid (\hat{B})_z \cap A \neq \emptyset\}$	Dilates the boundary of A . (I)
Opening	$A \circ B = (A \ominus B) \oplus B$	Smooths contours, breaks narrow isthmuses, and eliminates small islands and sharp peaks. (I)
Closing	$A \bullet B = (A \oplus B) \ominus B$	Smooths contours, fuses narrow breaks and long thin gulfs, and eliminates small holes. (I)
Hit-or-miss transform	$I \circledast B = \{z \mid (B)_z \subseteq I\}$	Finds instances of B in image I . B contains <i>both</i> foreground and background elements.
Boundary extraction	$\beta(A) = A - (A \ominus B)$	Set of points on the boundary of set A . (I)
Hole filling	$X_k = (X_{k-1} \oplus B) \cap I^c$ $k = 1, 2, 3, \dots$	Fills holes in A . X_0 is of same size as I , with a 1 in each hole and 0's elsewhere. (II)
Connected components	$X_k = (X_{k-1} \oplus B) \cap I$ $k = 1, 2, 3, \dots$	Finds connected components in I . X_0 is a set, the same size as I , with a 1 in each connected component and 0's elsewhere. (I)
Convex hull	$X_k^i = (X_{k-1}^i \circledast B^i) \cup X_{k-1}^i;$ $i = 1, 2, 3, 4 \quad k = 1, 2, 3, \dots$ $X_0^i = I; D^i = X_{conv}^i; C(A) = \bigcup_{i=1}^4 D^i$ X_{conv}^i means that $X_k^i = X_{k-1}^i$. (III)	Finds the convex hull, $C(A)$, of a set, A , of foreground pixels contained in image I .

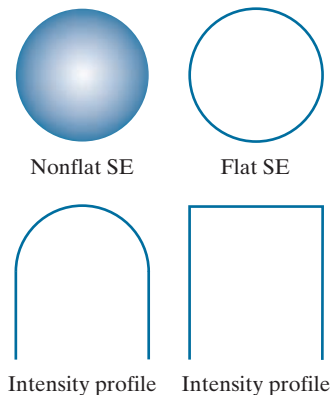
TABLE 9.1
(Continued)

Operation	Equation	Comments
Thinning	$A \otimes B = A - (A \otimes B)$ $= A \cap (A \otimes B)^c$ $A \otimes \{B\} =$ $\left(\left(\dots \left((A \otimes B^1) \otimes B^2 \right) \dots \right) \otimes B^n \right)$ $\{B\} = \{B^1, B^2, B^3, \dots, B^n\}$	Thins set A . The first two equations give the basic definition of thinning. The last two equations denote thinning by a sequence of structuring elements. This method is normally used in practice. (IV)
Thickening	$A \odot B = A \cup (A \otimes B)$ $A \odot \{B\} =$ $\left(\left(\dots \left((A \odot B^1) \odot B^2 \right) \dots \right) \odot B^n \right)$	Thickens set A using a sequence of structuring elements, as above. Uses (IV) with 0's and 1's reversed.
Skeletons	$S(A) = \bigcup_{k=0}^K S_k(A)$ $S_k(A) = (A \ominus kB)$ $- (A \ominus kB) \circ B$ Reconstruction of A: $A = \bigcup_{k=0}^K (S_k(A) \oplus kB)$	Finds the skeleton $S(A)$ of set A . The last equation indicates that A can be reconstructed from its skeleton subsets $S_k(A)$. K is the value of the iterative step after which the set A erodes to the empty set. The notation $(A \ominus kB)$ denotes the k th iteration of successive erosions of A by B . (I)
Pruning	$X_1 = A \otimes \{B\}$ $X_2 = \bigcup_{k=1}^8 (X_1 \otimes B^k)$ $X_3 = (X_2 \oplus H) \cap A$ $X_4 = X_1 \cup X_3$	X_4 is the result of pruning set A . The number of times that the first equation is applied to obtain X_1 must be specified. Structuring elements (V) are used for the first two equations. In the third equation H denotes structuring element. (I)
Geodesic dilation—size 1	$D_G^{(1)}(F) = (F \oplus B) \cap G$	F and G are called the <i>marker</i> and the <i>mask</i> images, respectively. (I)
Geodesic dilation—size n	$D_G^{(n)}(F) = D_G^{(1)}(D_G^{(n-1)}(F))$	Same comment as above.
Geodesic erosion—size 1	$E_G^{(1)}(F) = (F \ominus B) \cup G$	Same comment as above.
Geodesic erosion—size n	$E_G^{(n)}(F) = E_G^{(1)}(E_G^{(n-1)}(F))$	Same comment as above.
Morphological reconstruction by dilation	$R_G^D(F) = D_G^{(k)}(F)$	With k is such that $D_G^{(k)}(F) = D_G^{(k+1)}(F)$.

TABLE 9.1
(Continued)

Operation	Equation	Comments
Morphological reconstruction by erosion	$R_G^E(F) = E_G^{(k)}(F)$	With k such that $E_G^{(k)}(F) = E_G^{(k+1)}(F)$.
Opening by reconstruction	$O_R^{(n)}(F) = R_F^D(F \ominus nB)$	$F \ominus nB$ indicates n successive erosions by B , starting with F . The form of B is application-dependent.
Closing by reconstruction	$C_R^{(n)}(F) = R_F^E(F \oplus nB)$	$F \oplus nB$ indicates n successive dilations by B , starting with F . The form of B is application-dependent.
Hole filling	$H = [R_{I^c}^D(F)]^c$	H is equal to the input image I , but with all holes filled. See Eq. (9-45) for the definition of marker image F .
Border clearing	$X = I - R_I^D(F)$	X is equal to the input image I , but with all objects that touch (are connected to) the boundary removed. See Eq. (9-47) for the definition of marker image F .

a	b
c	d

FIGURE 9.36
Nonflat and flat structuring elements, and corresponding horizontal intensity profiles through their centers. All examples in this section are based on flat SEs.

where, in a manner similar to spatial correlation (see Section 3.4), x and y are incremented through all values required so that the origin of b visits every pixel in f . That is, to find the erosion of f by b , we place the origin of the structuring element at every pixel location in the image. The erosion at any location is determined by selecting the minimum value of f in the region coincident with b . For example, if b is a square structuring element of size 3×3 , obtaining the erosion at a point requires finding the minimum of the nine values of f contained in the 3×3 region spanned by b when its origin is at that point.

Similarly, the *grayscale dilation* of f by a flat structuring element b at any location (x, y) is defined as the *maximum* value of the image in the window spanned by $\hat{b}$ when the origin of $\hat{b}$ is at (x, y) . That is,

$$[f \oplus b](x, y) = \max_{(s, t) \in \hat{b}} \{f(x - s, y - t)\} \quad (9-50)$$

where we used the fact stated earlier that $\hat{b}(c, d) = b(-c, -d)$. The explanation of this equation is identical to the explanation in the previous paragraph, but using the maximum, rather than the minimum operation, and keeping in mind that the structuring element is reflected about its origin, which we take into account by using $(-s, -t)$ in the argument of the function. This is analogous to spatial convolution, as explained in Section 3.4.

EXAMPLE 9.9: Grayscale erosion and dilation.

Because grayscale erosion with a flat SE computes the minimum intensity value of f in every neighborhood of (x, y) coincident with b , we expect in general that an eroded grayscale image will be darker than the original, that the sizes (with respect to the size of the SE) of bright features will be reduced, and that the sizes of dark features will be increased. Figure 9.37(b) shows the erosion of Fig. 9.37(a) using a disk SE of unit height and a radius of 2 pixels. The effects just mentioned are clearly visible in the eroded image. For instance, note how the intensities of the small bright dots were reduced, making them barely visible in Fig. 9.37(b), while the dark features grew in thickness. The general background of the eroded image is slightly darker than the background of the original image.

Similarly, Fig. 9.37(c) is the result of dilation with the same SE. The effects are the opposite of using erosion. The bright features were thickened and the intensities of the darker features were reduced. In particular, the thin black connecting wires in the left, middle, and right bottom of Fig. 9.37(a) are barely visible in Fig. 9.37(c). The sizes of the dark dots were reduced as a result of dilation, but, unlike the eroded small white dots in Fig. 9.37(b), they still are easily visible in the dilated image. The reason is that the black dots were originally larger than the white dots with respect to the size of the SE. Finally, observe that the background of the dilated image is slightly lighter than that of Fig. 9.37(a).

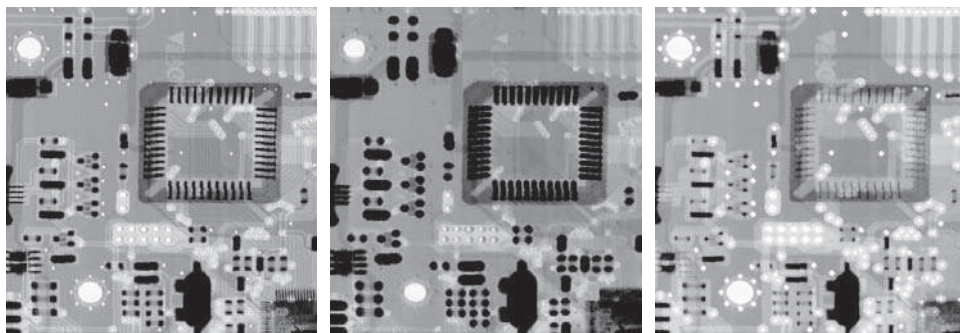
Nonflat SEs have grayscale values that vary over their domain of definition. The erosion of image f by nonflat structuring element, b_N , is defined as

$$[f \ominus b_N](x, y) = \min_{(s, t) \in b_N} \{f(x + s, y + t) - b_N(s, t)\} \quad (9-51)$$

a b c

FIGURE 9.37

(a) Gray-scale X-ray image of size 448×425 pixels. (b) Erosion using a flat disk SE with a radius of 2 pixels. (c) Dilation using the same SE. (Original image courtesy of Lixi, Inc.)



Here, we subtract values from f to determine the erosion at any point. Unlike Eq. (9-49), erosion using a nonflat SE is not bounded in general by the values of f , which can be problematic in interpreting results. Grayscale SEs are seldom used in practice because of this, the potential difficulties in selecting meaningful elements for b_N , and the added computational burden when compared with Eq. (9-49).

In a similar manner, dilation using a nonflat SE is defined as

$$[f \oplus b_N](x, y) = \max_{(s, t) \in \hat{b}_N} \{f(x - s, y - t) + \hat{b}_N(s, t)\} \quad (9-52)$$

The same comments made in the previous paragraph are applicable to dilation with nonflat SEs. When all the elements of b_N are constant (i.e., the SE is flat), Eqs. (9-51) and (9-52) reduce to Eqs. (9-49) and (9-50), respectively, within a scalar constant equal to the amplitude of the SE.

As in the binary case, grayscale erosion and dilation are duals with respect complementation and reflection; that is,

$$[f \ominus b]^c(x, y) = [f^c \oplus \hat{b}](x, y) \quad (9-53)$$

where $f^c(x, y) = -f(x, y)$ and $\hat{b}(x, y) = b(-x, -y)$. The same expression holds for nonflat structuring elements. Except as needed for clarity, we simplify the notation in the following discussion by suppressing the arguments of all functions, in which case the preceding equation is written as

$$(f \ominus b)^c = f^c \oplus \hat{b} \quad (9-54)$$

Similarly,

$$(f \oplus b)^c = f^c \ominus \hat{b} \quad (9-55)$$

Erosion and dilation by themselves are not particularly useful in grayscale image processing. As with their binary counterparts, these operations become powerful when used in combination to derive higher-level algorithms.

Although we deal with flat SEs in the following discussion, the concepts discussed are applicable also to nonflat structuring elements.

GRAYSCALE OPENING AND CLOSING

The expressions for opening and closing grayscale images have the same form as their binary counterparts. The *grayscale opening* of image f by structuring element b , denoted $f \circ b$, is

$$f \circ b = (f \ominus b) \oplus b \quad (9-56)$$

As before, opening is simply the erosion of f by b , followed by a dilation of the result by b . Similarly, the *grayscale closing* of f by b , denoted $f \bullet b$, is

$$f \bullet b = (f \oplus b) \ominus b \quad (9-57)$$

The opening and closing for grayscale images are duals with respect to complementation and SE reflection:

$$(f \bullet b)^c = f^c \circ \hat{b} \quad (9-58)$$

and

$$(f \circ b)^c = f^c \bullet \hat{b} \quad (9-59)$$

Because $f^c = -f$, we can write Eq. (9-58) as $-(f \bullet b) = (-f \circ b)$, and similarly for Eq. (9-59).

Opening and closing of grayscale images have a simple geometric interpretation. Suppose that an image function $f(x, y)$ is viewed as a 3-D surface; that is, its intensity values are interpreted as height values over the xy -plane, as in Fig. 2.18(a). Then the opening of f by b can be interpreted geometrically as pushing the structuring element up from below against the undersurface of f . At each location of the origin of b , the opening is the highest value reached by any part of b as it pushes up against the undersurface of f . The complete opening is then the set of all such values obtained by the origin of b visiting every (x, y) coordinate of f .

Figure 9.38 illustrates the concept in one dimension. Suppose the curve in Fig. 9.38(a) is the intensity profile along a single row of an image. Figure 9.38(b) shows a flat structuring element in several positions, pushed up against the bottom of the curve. The heavy curve in Fig. 9.38(c) is the complete opening. Because the structuring element is too large to fit completely inside the upward peaks of the curve, the tops of the peaks are clipped by the opening, with the amount removed being proportional to how far the structuring element was able to reach into the peak. In general, openings are used to remove small, bright details, while leaving the overall intensity levels and larger bright features relatively undisturbed.

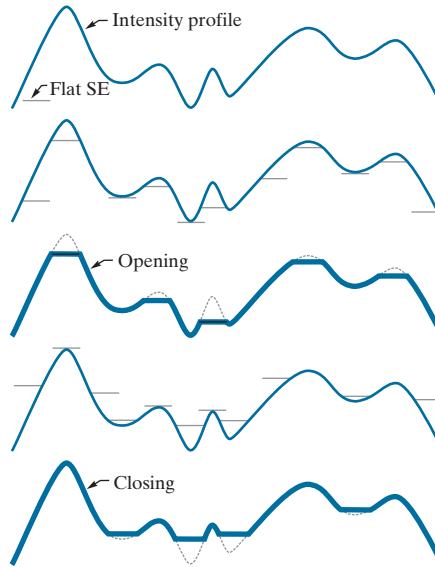
Figure 9.38(d) is a graphical illustration of closing. Observe that the structuring element is pushed down on top of the curve while being translated to all locations. The closing, shown in Fig. 9.38(e), is constructed by finding the lowest points reached by any part of the structuring element as it slides against the upper side of the curve. The grayscale opening satisfies the following properties:

a
b
c
d
e

FIGURE 9.38

Grayscale opening and closing in one dimension.

- (a) Original 1-D signal.
 (b) Flat structuring element pushed up underneath the signal.
 (c) Opening.
 (d) Flat structuring element pushed down along the top of the signal.
 (e) Closing.



- (a) $f \circ b \downarrow f$
 (b) If $f_1 \downarrow f_2$, then $(f_1 \circ b) \downarrow (f_2 \circ b)$
 (c) $(f \circ b) \circ b = f \circ b$

The notation $q \downarrow r$ is used to indicate that the domain of q is a subset of the domain of r , and also that $q(x, y) \leq r(x, y)$ for any (x, y) in the domain of q .

Similarly, the closing operation satisfies the following properties:

- (a) $f \downarrow f \bullet b$
 (b) If $f_1 \downarrow f_2$, then $(f_1 \bullet b) \downarrow (f_2 \bullet b)$
 (c) $(f \bullet b) \bullet b = f \bullet b$

The usefulness of these properties is similar to that of their binary counterparts.

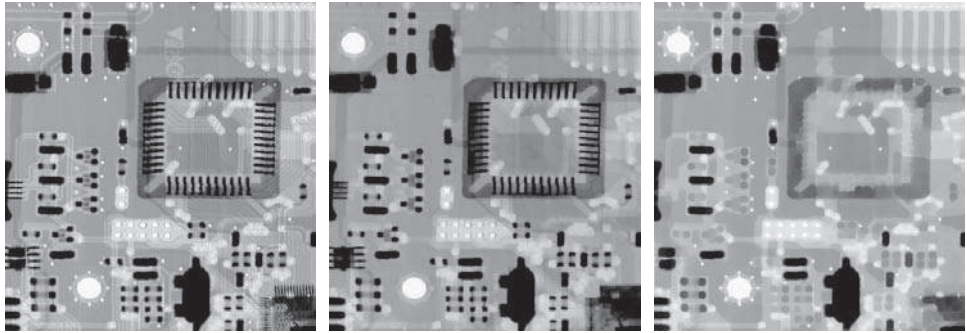
EXAMPLE 9.10: Grayscale opening and closing.

Figure 9.39 extends to 2-D the 1-D concepts illustrated in Fig. 9.38. Figure 9.39(a) is the same image we used in Example 9.9, and Fig. 9.39(b) is the opening obtained using a disk structuring element of unit height and radius of 3 pixels. As expected, the intensity of all bright features decreased, depending on the sizes of the features relative to the size of the SE. Comparing this figure with Fig. 9.37(b), we see that, unlike the result of erosion, opening had negligible effect on the dark features of the image, and the effect on the background was negligible. Similarly, Fig. 9.39(c) shows the closing of the image with a disk of radius 5 (the small round black dots are larger than the small white dots, so a larger disk was needed to achieve results comparable to the opening). In this image, the bright details and background were relatively unaffected, but the dark features were attenuated, with the degree of attenuation being dependent on the relative sizes of the features with respect to the SE.

a b c

FIGURE 9.39

(a) A grayscale X-ray image of size 448×425 pixels.
 (b) Opening using a disk SE with a radius of 3 pixels.
 (c) Closing using an SE of radius 5.



SOME BASIC GRAYSCALE MORPHOLOGICAL ALGORITHMS

Numerous grayscale morphological techniques are based on the grayscale morphological concepts introduced thus far. We illustrate some of these algorithms in the following discussion.

Morphological Smoothing

Because opening suppresses bright details smaller than the specified SE while leaving dark details relatively unaffected, and closing generally has the opposite effect, these two operations are used often in combination as *morphological filters* for image smoothing and noise removal. Consider Fig. 9.40(a), which shows an image of the Cygnus Loop supernova taken in the X-ray band (see Fig. 1.7 for details about this image). For purposes of the present discussion, suppose that the central light region is the object of interest, and that the smaller components are noise. Our objective is to remove the noise. Figure 9.40(b) shows the result of opening the original image with a flat disk of radius 1, then closing the opening with an SE of the same size. Figures 9.40(c) and (d) show the results of the same operation using SEs of radii 3 and 5, respectively. As expected, this sequence shows progressive removal of small components as a function of SE size. In the last result, we see that the noise has been almost eliminated. The noise components on the lower right side of the image could not be removed completely because their sizes are larger than the other image elements that were successfully removed.

The results in Fig. 9.40 are based on opening the original image, then closing the opening. A procedure used sometimes is to perform *alternating sequential filtering*, in which the opening-closing sequence starts with the original image, but subsequent steps perform the opening and closing on the results of the previous step. This type of filtering is useful in automated image analysis, in which results at each step are compared against a specified metric. This approach generally results in more blurring for the same size SE than the method illustrated in Fig. 9.40.

Morphological Gradient

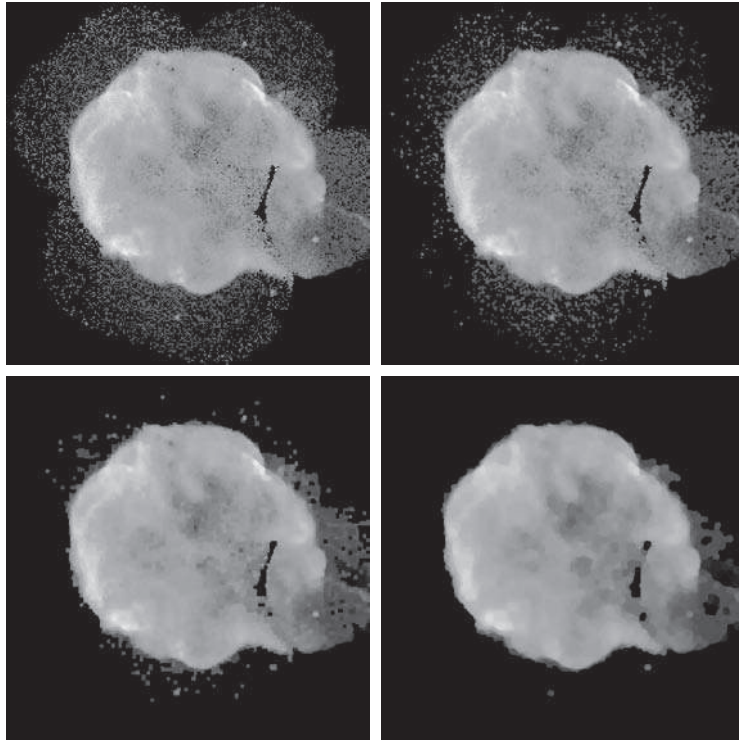
Dilation and erosion can be used in combination with image subtraction to obtain the morphological gradient, g , of a grayscale image f , as follows:

See Section 3.6 for a definition of the image gradient.

a	b
c	d

FIGURE 9.40

(a) 566×566 image of the Cygnus Loop supernova, taken in the X-ray band by NASA's Hubble Telescope.
 (b)–(d) Results of performing opening and closing sequences on the original image with disk structuring elements of radii, 1, 3, and 5, respectively. (Original image courtesy of NASA.)



$$g = (f \oplus b) - (f \ominus b) \quad (9-60)$$

where b is a suitable structuring element. The overall effect achieved by using this equation is that dilation thickens regions in an image, and erosion shrinks them. Their difference emphasizes the boundaries between regions. Homogenous areas are not affected (provided that the SE is not too large relative to the resolution of the image) so the subtraction operation tends to eliminate them. The net result is an image in which the edges are enhanced and the contribution of the homogeneous areas is suppressed, thus producing a “derivative-like” (gradient) effect.

Figure 9.41 shows an example. Figure 9.41(a) is a head CT scan, and the next two figures are the opening and closing with a 3×3 flat SE of 1's. Note the thickening and shrinking just mentioned. Figure 9.41(d) is the morphological gradient obtained using Eq. (9-60). As you can see, the boundaries between regions were clearly delineated, as expected of a 2-D derivative image.

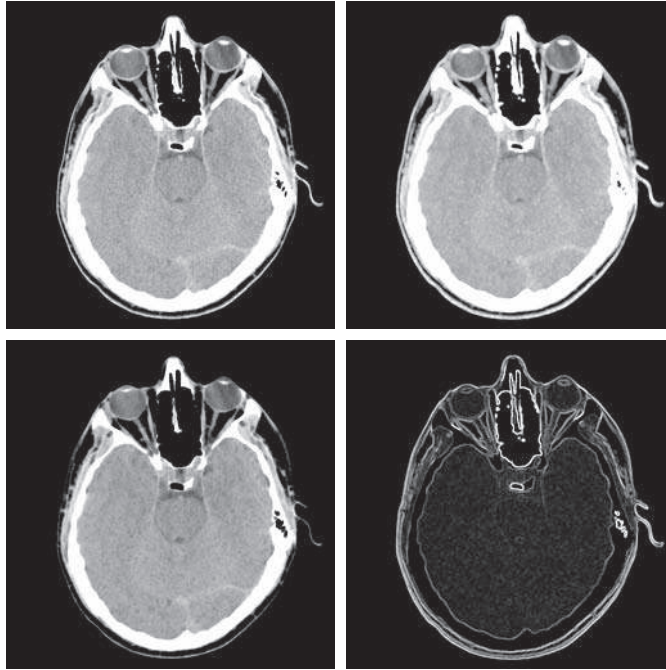
Top-Hat and Bottom-Hat Transformations

Combining image subtraction with openings and closings results in so-called top-hat and bottom-hat transformations. The *top-hat transformation* of a grayscale image f is defined as f minus its opening:

a	b
c	d

FIGURE 9.41

(a) 512×512 image of a head CT scan.
 (b) Dilation.
 (c) Erosion.
 (d) Morphological gradient, computed as the difference between (b) and (c). (Original image courtesy of Dr. David R. Pickens, Vanderbilt University.)



$$T_{\text{hat}}(f) = f - (f \circ b) \quad (9-61)$$

Similarly, the *bottom-hat transformation* of f is defined as the closing of f minus f :

$$B_{\text{hat}}(f) = (f \bullet b) - f \quad (9-62)$$

One of the principal applications of these transformations is in removing objects from an image by using a structuring element in the opening or closing operation that does not fit the objects to be removed. The difference operation then yields an image in which only the removed components remain. The top-hat transformation is used for light objects on a dark background, and the bottom-hat transformation is used for the opposite situation. For this reason, the names *white top-hat* and *black top-hat*, respectively, are used frequently when referring to these two transformations.

An important use of top-hat transformations is for correcting the effects of non-uniform illumination. As you will learn in Chapter 10, proper (uniform) illumination plays a central role in being able to extract objects from the background in an image. This process is fundamental in automated image analysis, and is often used in conjunction with thresholding, as you will learn in Chapter 10.

To illustrate, consider Fig. 9.42(a), which shows an image of grains of rice. This image was obtained under nonuniform lighting, as evidenced by the darker area in the bottom rightmost part of the image. Figure 9.42(b) shows the result of thresholding using Otsu's method, an optimal thresholding method to be discussed in Section 10.3. The net result of nonuniform illumination was to cause segmentation

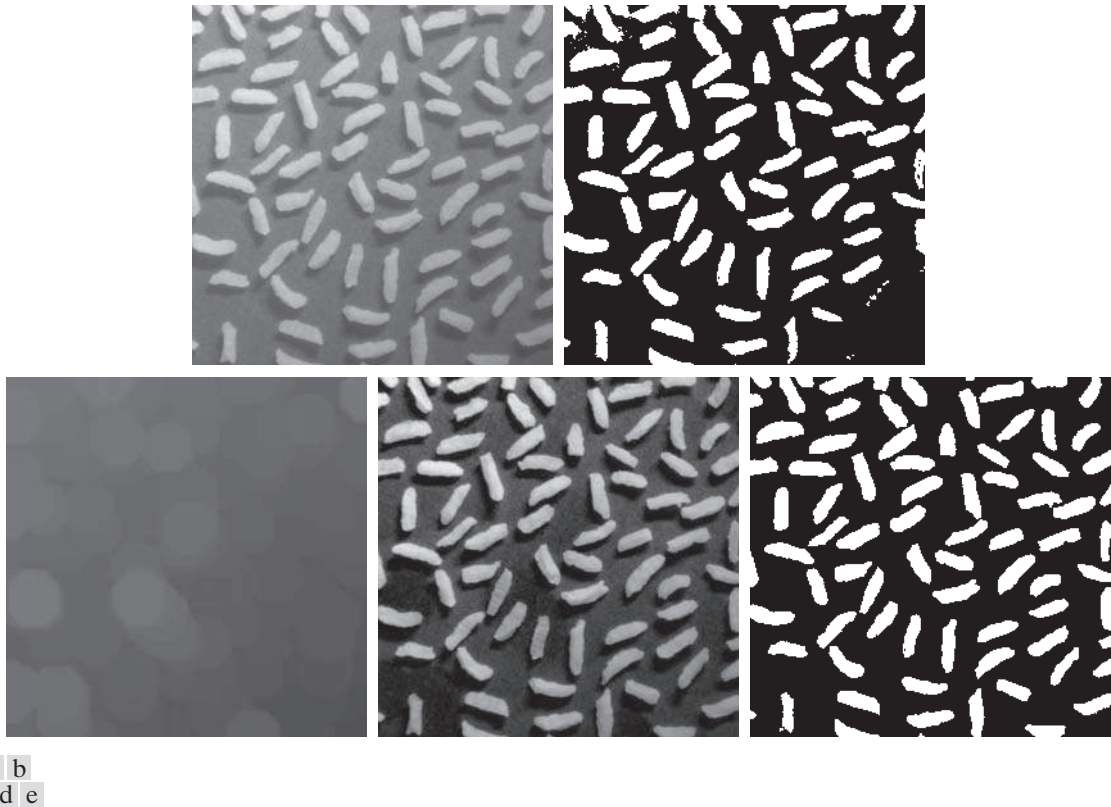


FIGURE 9.42 Using the top-hat transformation for *shading correction*. (a) Original image of size 600×600 pixels. (b) Thresholded image. (c) Image opened using a disk SE of radius 40. (d) Top-hat transformation (the image minus its opening). (e) Thresholded top-hat image.

errors in the dark area (several grains of rice were not extracted from the background), as well as in the top left part of the image, where parts of the background were interpreted as rice. Figure 9.42(c) shows the opening of the image with a disk of radius 40. This SE was large enough so that it would not fit in any of the objects. As a result, the objects were eliminated, leaving only an approximation of the background. The shading pattern is clear in this image. By subtracting this image from the original (i.e., by applying a top-hat transformation), the background should become more uniform. This is indeed the case, as Fig. 9.42(d) shows. The background is not perfectly uniform, but the differences between light and dark extremes are less, and this was enough to yield a correct thresholding result, in which all the rice grains were properly extracted using Otsu's method, as Fig. 9.42(e) shows.

Granulometry

In the context of this discussion, *granulometry* is a field that deals with determining the size distribution of particles in an image. Particles seldom are neatly separated,

which makes counting based on identifying individual particles a difficult task. Morphology can be used to estimate particle size distribution indirectly, without having to identify and measure individual particles.

The approach is simple. With particles having regular shapes that are lighter than the background, the method consists of applying openings with SEs of increasing sizes. The basic idea is that opening operations of a particular size should have the most effect on regions of the input image that contain particles of similar size. For each image resulting from an opening, we compute the sum of the pixel values. This sum, called the *surface area*, decreases as a function of increasing SE size because, as we discussed earlier, openings decrease the intensity of light features in an image. This procedure yields a 1-D array each element of which is the sum of the pixels in the opening for the size SE corresponding to that location in the array. To emphasize changes between successive openings, we compute the difference between adjacent elements of the 1-D array. If the differences are plotted, the peaks in the plot are an indication of the predominant size distributions of the particles in the image.

As an example, consider the image of wood dowel plugs of two dominant sizes in Fig. 9.43(a). The wood grain in the dowels is likely to introduce variations in the openings, so smoothing is a sensible preprocessing step. Figure Fig. 9.43(b) shows the image smoothed using the morphological smoothing filter discussed earlier, with a disk of radius 5. Figures 9.43(c) through (f) show image openings with disks of radii 10, 20, 25, and 30, respectively. Note in Fig. 9.43(d) that the intensity contribution due to the small dowels has been almost eliminated. In Fig. 9.43(e) the contribution of the large dowels has been reduced significantly, and in Fig. 9.43(f) even more so. Observe in Fig. 9.43(e) that the large dowel near the top right of the image is much darker than the others because its size is smaller than other larger dowels. This would be useful information if we had been attempting to detect defective dowels.

Figure 9.44 shows a plot of the difference array. As mentioned previously, we expect significant differences (peaks in the plot) around radii at which the SE is

a	b	c
d	e	f

FIGURE 9.43

(a) 531×675 image of wood dowels.

(b) Smoothed image.

(c)–(f) Openings of (b) with disks of radii equal to 10, 20, 25, and 30 pixels, respectively.

(Original image courtesy of Dr. Steve Eddins, MathWorks, Inc.)

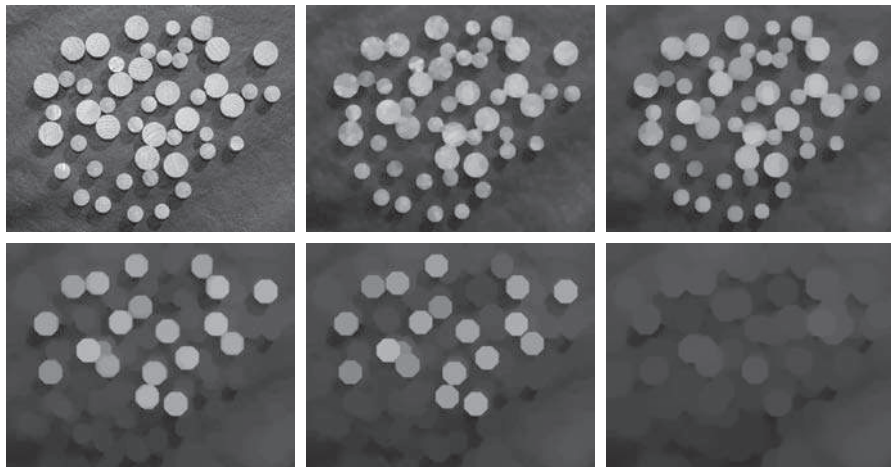
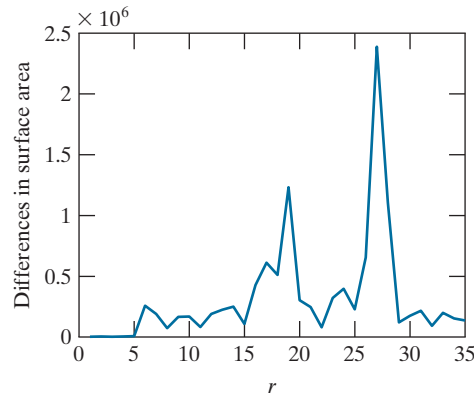


FIGURE 9.44

Differences in surface area as a function of SE disk radius, r . The two peaks indicate that there are two dominant particle sizes in the image.



large enough to encompass a set of particles of approximately the same diameter. The result in Fig. 9.44 has two distinct peaks, clearly indicating the presence of two dominant object sizes in the image.

Textural Segmentation

Figure 9.45(a) shows a noisy image of dark blobs superimposed on a light background. The image has two textural regions: a region composed of large blobs on the right and a region on the left composed of smaller blobs. The objective is to find a boundary between the two regions based on their textural content, which in this case is determined by the sizes and spatial distribution of the blobs (we discuss texture in Chapter 11). The process of partitioning an image into regions is called *segmentation*, which is the topic of Chapter 10.

The objects of interest are darker than the background, and we know that if we close the image with a structuring element larger than the small blobs, these blobs will be removed. The result in Fig. 9.45(b), obtained by closing the input image using a disk with a radius of 30 pixels, shows that indeed this is the case. (The radius of the smaller blobs is approximately 25 pixels.) So, at this point, we have an image with large, dark blobs on a light background. If we open this image with a structuring element that is large relative to the separation between these blobs, the net result should be an image in which the light patches between the blobs are removed, leaving the dark blobs, and also the now dark patches between these blobs. Figure 9.45(c) shows the result, obtained using a disk of radius 60.

Performing a morphological gradient on this image with, say, a 3×3 SE of 1's, will give us the boundary between the two regions. Figure 9.45(d) shows the boundary obtained from the morphological gradient operation, superimposed on the original image. All pixels to the right of this boundary are said to belong to the texture region characterized by large blobs, and conversely for the pixels on the left of the boundary. You will find it instructive to work through this example in more detail using the graphical analogy for opening and closing illustrated in Fig. 9.38.

a	b
c	d

FIGURE 9.45

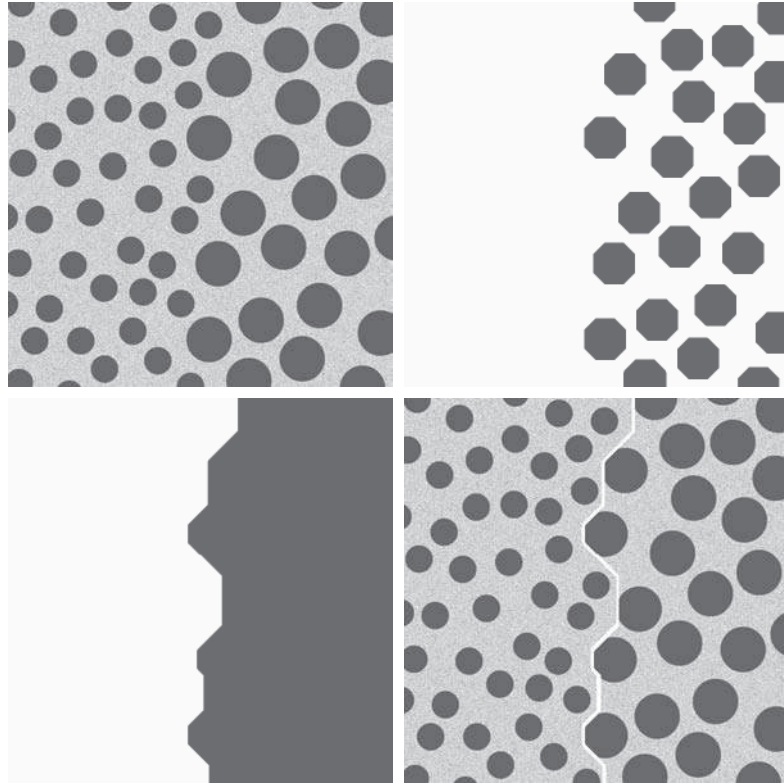
Textural segmentation.

(a) A 600×600 image consisting of two types of blobs.

(b) Image with small blobs removed by closing (a).

(c) Image with light patches between large blobs removed by opening (b).

(d) Original image with boundary between the two regions in (c) superimposed. The boundary was obtained using a morphological gradient.



As mentioned earlier, it is understood that f and g are functions of x and y . We omit the coordinates to simplify the notation.

GRAYSCALE MORPHOLOGICAL RECONSTRUCTION

Grayscale morphological reconstruction is defined in the same manner introduced in Section 9.6 for binary images. Let f and g denote the *marker* and *mask* images, respectively. We assume that both are grayscale images of the same size and that $f \leq g$, meaning that the intensity of f at any point in the image is less than the intensity of g at that point. The *geodesic dilation of size 1 of f with respect to g* is defined as

$$D_g^{(1)}(f) = (f \oplus b) \wedge g \quad (9-63)$$

where $\wedge$ denotes the *point-wise minimum operator*, and b is a suitable structuring element. We see that the geodesic dilation of size 1 is obtained by first computing the dilation of f by b , then selecting the minimum between the result and g at every point (x, y) . The dilation is given by Eq. (9-50) if b is a flat SE, or by Eq. (9-52) if it is not.

The *geodesic dilation of size n of f with respect to g* is defined as

$$D_g^{(n)}(f) = D_g^{(1)}(D_g^{(n-1)}(f)) \quad (9-64)$$

with $D_g^{(0)}(f) = f$.

See Problem 9.33 for a list of dual relationships between expressions in this section.

Similarly, the *geodesic erosion of size 1 of f with respect to g* is defined as

$$E_g^{(1)}(f) = (f \ominus b) \vee g \quad (9-65)$$

where $\vee$ denotes the *point-wise maximum operator*. The *geodesic erosion of size n* is defined as

$$E_g^{(n)}(f) = E_g^{(1)}(E_g^{(n-1)}(f)) \quad (9-66)$$

with $E_g^{(0)}(f) = f$.

The *morphological reconstruction by dilation* of a *grayscale mask image, g* , by a *grayscale marker image, f* , denoted by $R_g^{(D)}(f)$, is defined as the geodesic dilation of f with respect to g , iterated until stability is reached; that is,

$$R_g^D(f) = D_g^{(k)}(f) \quad (9-67)$$

with k such that $D_g^{(k)}(f) = D_g^{(k+1)}(f)$. The *morphological reconstruction by erosion* of f by g , denoted by $R_g^E(f)$, is similarly defined as

$$R_g^E(f) = E_g^{(k)}(f) \quad (9-68)$$

with k such that $E_g^{(k)}(f) = E_g^{(k+1)}(f)$.

As in the binary case, opening by reconstruction of grayscale images first erodes the input image and uses it as a marker, and uses the image itself as the mask. The *opening by reconstruction of size n* of an image f is defined as the reconstruction by dilation of the erosion of size n of f with respect to f ; that is,

$$O_R^{(n)}(f) = R_f^D(f \ominus nb) \quad (9-69)$$

where $f \ominus nb$ denotes n successive erosions by b , starting with f , as explained in connection with Eq. (9-30) (note that f itself is used as the mask). Recall also from the discussion of Eq. (9-44) for binary images that the objective of opening by reconstruction is to preserve the shape of the image components that remain after erosion.

Similarly, the *closing by reconstruction of size n* of an image f is defined as the reconstruction by erosion of the dilation of size n of f with respect to f ; that is,

$$C_R^{(n)}(f) = R_f^E(f \oplus nb) \quad (9-70)$$

where $f \oplus nb$ denotes n successive dilations by b , starting with f . Because of duality, the closing by reconstruction of an image can be obtained by complementing the image, obtaining the opening by reconstruction, and complementing the result. Finally, as the following example shows, a useful technique called *top-hat by reconstruction* consists of subtracting from an image its opening by reconstruction.

EXAMPLE 9.11: Using grayscale morphological reconstruction to flatten a complex background.

In this example, we illustrate the use of grayscale reconstruction in several steps. The objective is to normalize the irregular background of the image in Fig. 9.46(a), leaving only text on a background of

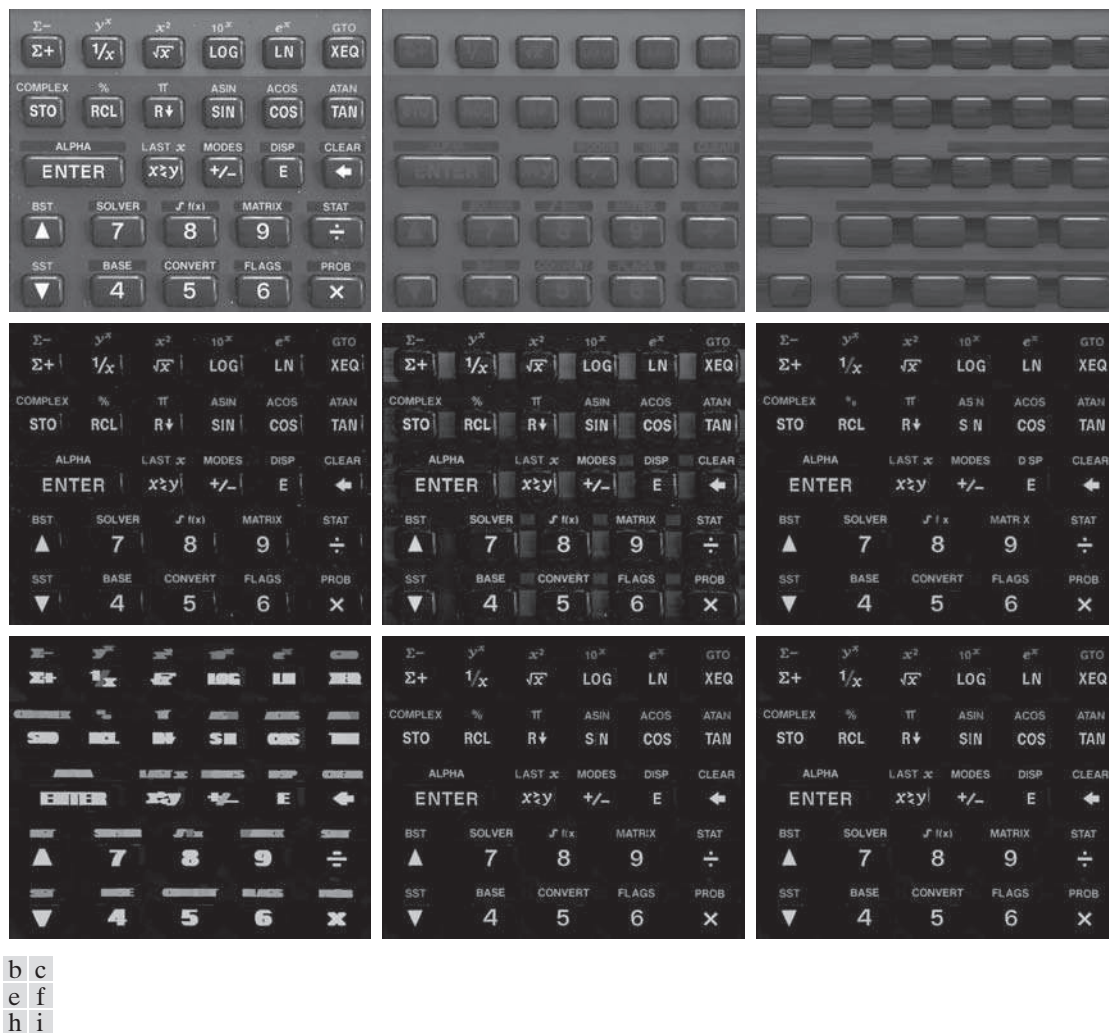


FIGURE 9.46 (a) Original image of size 1134×1360 pixels. (b) Opening by reconstruction of (a), using a structuring element consisting of a horizontal line 71 pixels long in the erosion. (c) Opening of (a) using the same SE. (d) Top-hat by reconstruction. (e) Result of applying just a top-hat transformation. (f) Opening by reconstruction of (d), using a horizontal line 11 pixels long. (g) Dilation of (f) using a horizontal line 21 pixels long. (h) Minimum of (d) and (g). (i) Final reconstruction result. (Images courtesy of Dr. Steve Eddins, MathWorks, Inc.)

constant intensity. The solution of this problem is a good illustration of the power of grayscale morphology. We begin by suppressing the horizontal reflection on the top of the keys. The reflections are wider than any single character in the image, so we should be able to suppress them by performing an opening by reconstruction using a long horizontal line in the erosion operation. This operation will yield the background containing the keys and their reflections. Subtracting this from the original image (i.e., performing a top-hat by reconstruction) will eliminate the horizontal reflections and variations in background from the original image.

Figure 9.46(b) shows the result of opening by reconstruction of the original image using a horizontal line of size 1×71 pixels for the SE in the erosion operation. We could have used an opening to remove the characters, but the resulting background would not have been as uniform, as Fig. 9.46(c) shows (compare the regions between the keys in the two images). Figure 9.46(d) shows the result of subtracting Fig. 9.46(b) from Fig. 9.46(a). As expected, the horizontal reflections and variations in background were suppressed. For comparison, Fig. 9.46(e) shows the result of performing just a top-hat transformation (i.e., subtracting the “standard” opening from the image). As expected from the characteristics of the background in Fig. 9.46(c), the background in Fig. 9.46(e) is not nearly as uniform as in Fig. 9.46(d).

The next step is to remove the vertical reflections from the edges of the keys, visible in Fig. 9.46(d). We can do this by performing an opening by reconstruction with a line SE whose width is approximately equal to the reflections (about 11 pixels in this case). Figure 9.46(f) shows the result of performing this operation on Fig. 9.46(d). The vertical reflections were suppressed, but so were thin, vertical strokes that are valid characters (for example, the I in SIN), so we have to find a way to restore the latter. The suppressed characters are very close to the other characters so, if we dilate the remaining characters horizontally, the dilated characters will overlap the area previously occupied by the suppressed characters. Figure 9.46(g), obtained by dilating Fig. 9.46(f) with a line SE of size 1×21 elements, shows that indeed this is case.

All that remains at this point is to restore the suppressed characters. Consider an image formed as the point-wise minimum between the dilated image in Fig. 9.46(g) and the top-hat by reconstruction in Fig. 9.46(d). Figure 9.46(h) shows the minimum image (although this result appears to be close to our objective, note that the I in SIN is still missing). By using this image as a marker and the dilated image as the mask in grayscale reconstruction [Eq. (9-67)], we obtained the final result in Fig. 9.46(i). This image shows that all characters were properly extracted from the original, irregular background, including the background of the keys. The background in Fig. 9.46(i) is uniform throughout.

Summary, References, and Further Reading

The morphological concepts and techniques introduced in this chapter constitute a powerful set of tools for extracting features of interest in an image. One of the most appealing aspects of morphological image processing is the extensive set-theoretic foundation from which morphological techniques have evolved. A significant advantage in terms of implementation is that dilation and erosion are primitive operations, which are the basis for a broad class of morphological algorithms. As will be shown in the following chapter, morphology can be used as the basis for developing image segmentation procedures with numerous applications. As we will discuss in Chapter 11, morphological techniques also play a major role in procedures for image feature extraction.

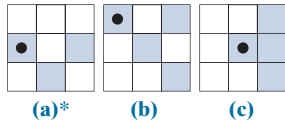
The book by Serra [1982] is a fundamental reference on morphological image processing. See also Serra [1988], Giardina and Dougherty [1988], and Haralick and Shapiro [1992]. For an overview of both binary and gray-scale morphology, see Basart and Gonzalez [1992] and Basart et al. [1992]. This set of references provides ample basic background for the material covered in Sections 9.1 through 9.4. For a good overview of the material in Sections 9.5 and 9.6, see the book by Soille [2003].

Important issues of implementing morphological algorithms such as the ones given in Section 9.5 and 9.6 are exemplified in the papers by Jones and Svalbe [1994], Sussner and Ritter [1997], and Shaked and Bruckstein [1998]. A paper by Vincent [1993] is especially important in terms of practical details for implementing gray-scale morphological algorithms. For additional reading on the theory and applications of morphological image processing, see the books by Goutsias and Bloomberg [2000], and by Beyer et al. [2016]. To get an idea of the state of the art in fast computer implementation of morphological algorithms, see Thurley and Danell [2012]. For details on the software aspects of many of the examples in this chapter, see Gonzalez, Woods, and Eddins [2009].

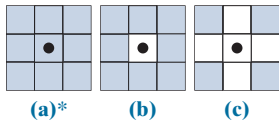
Problems

Solutions to the problems marked with an asterisk (*) are in the DIP4E Student Support Package (consult the book website: www.ImageProcessingPlace.com).

- 9.1** Find the reflection, $\hat{B}$, of each of the following structuring elements. The dot indicates the origin of the SE.



- 9.2** Sketch the result of eroding Fig. 9.3(a) with each of the following structuring elements.



- 9.3*** Erosion of a set A by structuring element B is a subset of A , provided that the origin of B lies within B . Give an example in which the erosion $A \ominus B$ lies outside, or partially outside, A .

- 9.4** Let B be a structuring element containing a single point, valued 1, and let A be a set of foreground pixels.

(a)* What do you think would happen if we erode A by B ?

(b) What do you think would happen if we dilate A by B ?

- 9.5** You are given a “black-box” function that computes erosion. You are told that this function automatically pads the input image with a border whose width is the thinnest border possible, as determined by the dimensions of the structuring element (e.g., for a 3×3 structuring element the border would be one pixel wide). However, you are not told whether the padding is composed of background (0) or foreground (1) values. Propose an experiment for answering this question.

- 9.6** Do the following:

(a)* Dilate Fig. 9.3(a) using the structuring element in figure (a) of Problem 9.2.

(b) Repeat (a) using the structuring element in figure (b).

(c) Repeat (a) using the structuring element in figure (c).

- 9.7** Dilation of a set A by structuring element B is the set of locations of the origin of B such that A contains at least one (foreground) element of B . Give an example in which the dilation of A by B lies completely outside of A . (Hint: Let A and B be disks of different radii.)

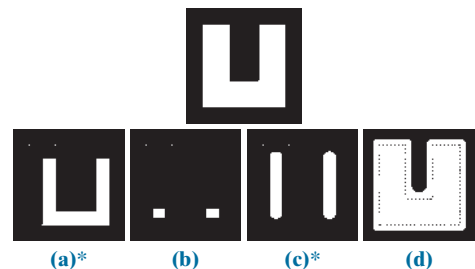
- 9.8** With reference to the image at the top of the figure shown below, answer the following:

(a)* Give the structuring element and morphological operation(s) that produced image (a). Show the origin of the structuring element. The dashed lines denote the boundary of the original object and are shown for reference; they are not part of the result. (The white elements are foreground pixels.)

(b) Repeat part (a) for the output shown in image (b).

(c)* Repeat part (a) for the output shown in image (c).

(d) Repeat part (a) for the solution shown in figure (d). Note that in image (d) all corners are rounded.

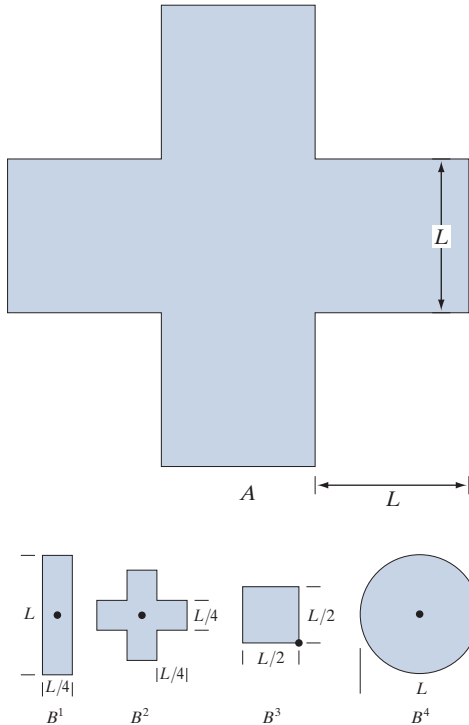


- 9.9** Let A denote the set shown shaded in the following figure, and refer to the structuring elements shown (the black dots denote the origin). Sketch the result of the following operations:

(a)* $(A \ominus B^4) \oplus B^2$.

(b) $(A \ominus B^1) \oplus B^3$.

(c) $(A \oplus B^1) \oplus B^3$.



9.13 Do the following:

- (a) Show that the definition of erosion given in Problem 9.12 is equivalent to yet another definition of erosion:

$$A \ominus B = \bigcap_{b \in B} (A)_{-b}$$

(If $-b$ is replaced with b , this expression is called the *Minkowsky subtraction* of two sets.)

- (b)* Show that the expression in (a) is equivalent to the definition in Eq. (9-3).

9.14* An alternative definition of dilation is

$$A \oplus B = \{w \in Z^2 \mid w = a + b, \text{ for some } a \in A \text{ and } b \in B\}$$

Show that this definition and the definition in Eq. (9-6) are equivalent.

9.15 Do the following:

- (a) Show that the definition of dilation given in Problem 9.14 is equivalent to yet another definition of dilation:

$$A \oplus B = \bigcup_{b \in B} (A)_b$$

(This expression is called the *Minkowsky addition* of two sets.)

- (b)* Show that the expression in (a) is equivalent also to the definition in Eq. (9-6).

9.16 Prove the validity of the duality expression given in Eq. (9-9).

9.17 Answer the following:

- (a)* The curved portions the black border of Fig. 9.8(d) delineate the opening of set A in Fig. 9.8(a), but those curved segments are not part of the boundary of A . Are the black straight-line portions in (d) part of the boundary of A ? Explain.
- (b) The curved portions the black border of Fig. 9.9(d) delineate the closing of set A in Fig. 9.9(a), but those curved segments are not part of the boundary of A . Are the black straight line portions of the boundary in (d) part of the boundary of A ? Explain.

9.10 Be specific in answering the following:

- (a)* What is the limiting effect of repeatedly dilating a set of foreground pixels in an image? Assume that a trivial (one point) structuring element is not used.
- (b) What is the smallest set from which you can start in order for your answer in (a) to hold?

9.11 Be specific in answering the following:

- (a) What is the limiting effect of repeatedly eroding a set of foreground pixels in an image? Assume that a trivial (one point) structuring element is not used.
- (b) What is the smallest set of foreground pixels from which you can start in order for your answer in (a) to hold?

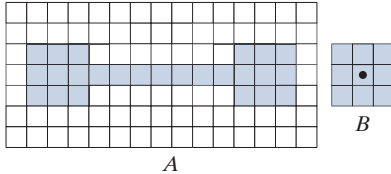
9.12* An alternative definition of erosion is

$$A \ominus B = \{w \in Z^2 \mid w + b \in A \text{ for every } b \in B\}$$

Show that this definition is equivalent to the definition in Eq. (9-3).

9.18 Show all intermediate steps of your computations for the following:

- (a)* Obtain the opening of the figure below using a 3×3 SE of 1's. Do all operations manually.
 (b) Repeat (a) for the closing operation.



9.19 A is a solid rectangle of 1's of size $M \times N$ with a 1-pixel border of 0's, and m and n below are odd integers. Discuss what the result will be in each case.

- (a)* A is opened with a structuring element of 1's of size $m \times n$.
 (b) A is closed with a structuring element of 1's of size $m \times n$.

9.20 Show the validity of the following duality expressions [these are Eqs. (9-14) and (9-15)]:

- (a)* $(A \circ B)^c = A^c \bullet \hat{B}$.
 (b) $(A \bullet B)^c = A^c \circ \hat{B}$.

9.21 Show the validity of the following expressions:

- (a)* $A \circ B$ is a subset of A . You may assume that Eq. (9-12) is valid. (Hint: Start with this equation and Fig. 9.8.)
 (b)* If C is a subset of D , then $C \circ B$ is a subset of $D \circ B$. [Hint: Start with Eq. (9-12).]
 (c) $(A \circ B) \circ B = A \circ B$. [Hint: Start with the definition of opening.]

9.22 Show the validity of the following expressions. (Hint: Study the solution to Problem 9.21.)

- (a) A is a subset of $A \bullet B$.
 (b) If C is a subset of D , then $C \bullet B$ is a subset of $D \bullet B$.
 (c) $(A \bullet B) \bullet B = A \bullet B$.

9.23 Refer to the image and the disk structuring element shown in the lower right of the image. Sketch what the sets C , D , E , and F would look like for the following sequence of operations: $C = A \ominus B$; $D = C \oplus B$; $E = D \oplus B$; and $F = E \ominus B$. Set A consists of all the foreground pixels (white),

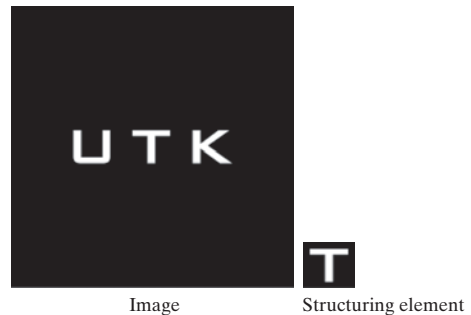
except the structuring element, B , which you may assume is just large enough to encompass any of the random elements in the image. Note that the sequence of operations above is simply the opening of A by B followed by a closing of the result by B .



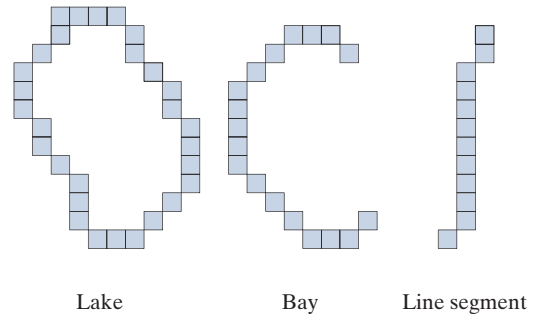
9.24* Assume that SE B_2 in Fig. 9.12 has a border of foreground pixels that is more than one pixel wide. Assuming that all four sides of the border are the same, what is the maximum width of a border we can use around B_2 before the solution shown in Fig. 9.12(f) fails?

9.25 We mentioned when discussing Fig. 9.12(e) that the image had been cropped for consistency. Assume that Fig. 9.12(b) was padded with the minimum border required to encompass the maximum excursions of B_2 after which no further changes would occur in the erosion. What did Fig. 9.12(e) look like before it was cropped?

9.26 Sketch the result of applying the hit-or-miss transform to the image below, using the SE shown. Indicate clearly the origin and border you selected for the structuring element.



- 9.27*** Give the foundation of an algorithm for converting an 8-connected, closed curve to a 4-connected curve (see Section 2.5 regarding connectivity). The input is a binary image, I , in which the curve consists of 1-valued pixels embedded in a background of 0's. The output should be a binary image also, containing the new curve. You may assume that the curve is fully connected, is one pixel thick, and has no branches. You do not need to (but you may) state the algorithm in a step-by-step manner. An overall plan containing all the information needed to implement a working algorithm is sufficient.
- 9.28** Give the foundation of an algorithm for converting a 4-connected closed curve to a curve containing *only* 8-connected pixels (see Section 2.5 regarding connectivity). The input is a binary image, I , in which the curve consists of 1-valued pixels embedded in a background of 0's. The output should be a binary image also, containing the new curve. You may assume that the curve is fully connected, it is one-pixel-wide, and has no branches. You do not need to (but you may) state the algorithm in a step-by-step manner. An overall plan containing all the information needed to implement a working algorithm is sufficient.
- 9.29** Give the foundation of an algorithm for converting an 8-connected closed curve to an m -connected curve (see Section 2.5 regarding connectivity). The input is a binary image, I , in which the curve consists of 1-valued pixels embedded in a background of 0's. The output should be a binary image also, containing the new curve. You may assume that the curve is fully connected, it is one-pixel-wide, and has no branches. You do not need to (but you may) state the algorithm in a step-by-step manner. An overall plan containing all the information needed to implement a working algorithm is sufficient.
- 9.30*** Three curve types (lake, bay, and line segment) useful for differentiating thinned objects in an image are shown in the following figure. Develop a morphological/logical algorithm for differentiating between these shapes. The input to your algorithm would be one of these three curves. The output must be the type of the input. You may assume that the curves are 1 pixel thick and are fully connected. They can appear in any orientation.



- 9.31** Write Eq. (9-18) in terms of a dilation, instead of an erosion, of A . (*Hint:* Take a look at the definition of set difference in Eq. (2-40) and then consider the duality relationship between erosion and dilation.)
- 9.32** Answer the following:
- (a)* Discuss the effect of using the structuring element in Fig. 9.17(c) for boundary extraction, instead of the element in Fig. 9.15(b).
 - (b) What would be the effect of using a 3×3 structuring element composed of all 1's in the hole filling algorithm of Eq. (9-19), instead of the structuring element in Fig. 9.17(c)?
- 9.33** Discuss what you would expect the result to be in each of the following cases:
- (a)* The starting point of the hole filling algorithm of Eq. (9-19) is a point *on* the outer boundary of the object containing the hole.
 - (b) The starting point in the hole filling algorithm is *outside* of the boundary (i.e., the starting point is a background pixel).
- 9.34** Sketch the convex hull of the large figure in Problem 9.9. Assume that $L = 3$ pixels.
- 9.35** Obtain the convex deficiency of set A shown in Fig. 9.21(b). Use the convex hull in Fig. 9.22(a).
- 9.36** Do the following:
- (a)* Propose a method using any of the methods developed in this chapter for automating the example in Fig. 9.18. You may assume that the spheres do not touch each other and that none touch the border of the image.
 - (b) Repeat (a), but allowing the spheres to touch in arbitrary ways, including the border.

9.37* The algorithm for extracting connected components discussed in Section 9.5 requires that a point be known in each connected component in order to extract them all. Suppose that you are given a binary image containing an arbitrary (unknown) number of connected components. Propose a completely automated procedure for extracting all connected components. Assume that points belonging to connected components are labeled 1 and background points are labeled 0.

9.38 Give an expression based on reconstruction by dilation capable of extracting all the holes in a binary image.

9.39 With reference to the hole-filling algorithm in Eqs. (9-45) and (9-46):

- (a)* Explain what would happen if all border points of I are 1 (foreground).
- (b) If the result in (a) gives the result that you would expect, explain why. If it does not, explain how you would modify the algorithm so that it works as expected.

9.40* As explained in Eqs. (9-44) and (9-69), opening by reconstruction preserves the shape of the image components that remain after erosion. What does closing by reconstruction do?

9.41 Show that geodesic erosion and dilation (Section 9.6) are duals with respect to set complementation. That is, assuming that the structuring element is symmetric about its origin, show that:

- (a)* $E_G^{(n)}(F) = \left[D_G^{(1)} \left[D_G^{(n-1)}(F^c) \right] \right]^c$ and, conversely, that
- (b) $D_G^{(n)}(F) = \left[E_G^{(1)} \left[E_G^{(n-1)}(F^c) \right] \right]^c$.

(Hint: Use proof by induction.)

9.42 Show that reconstruction by dilation and reconstruction by erosion (Section 9.6) are duals with respect to set complementation. That is, assuming that the structuring element is symmetric about its origin, show that $R_G^D(F) = \left[R_G^E(F^c) \right]^c$ and, conversely, that $R_G^E(F) = \left[R_G^D(F^c) \right]^c$. (Hint: Consider using the results from Problem 9.41.)

9.43 Show that:

- (a)* $(F \ominus nB)^c = F^c \oplus n\hat{B}$, where $F \ominus nB$ indicates n successive erosions, starting with B ; and similarly, that
- (b) $(F \oplus nB)^c = F^c \ominus n\hat{B}$.

9.44 Show the validity of the following binary morpho-

logical expressions. You may assume that the structuring element is symmetric about its origin.

(a)* $O_R^{(n)}(F) = \left[C_R^{(n)}(F^c) \right]^c$.

(b) $C_R^{(n)}(F) = \left[O_R^{(n)}(F^c) \right]^c$.

9.45 Prove the validity of the following grayscale morphological expressions. Recall from the discussion in Section 9.8 that $f^c(x, y) = -f(x, y)$ and that $\hat{b}(x, y) = b(-x, -y)$.

(a)* $(f \ominus b) = f^c \oplus b$.

(b) $(f \oplus b)^c = f^c \ominus \hat{b}$.

(c) $(f \cdot b)^c = f^c \circ \hat{b}$.

(d)* $(f \circ b)^c = f^c \cdot \hat{b}$.

9.46 Prove the validity of the following grayscale morphological expressions. Recall that $f^c(x, y) = -f(x, y)$ and that $\hat{b}(x, y) = b(-x, -y)$. (Hint: Use proof by induction.)

(a)* $D_g^{(n)}(f) = \left[E_g^{(1)} \left[E_g^{(n-1)}(f^c) \right] \right]^c$. Assume a symmetric structuring element.

(b) $E_g^{(n)}(f) = \left[D_g^{(1)} \left[D_g^{(n-1)}(f^c) \right] \right]^c$. Assume a symmetric structuring element.

9.47 Prove the validity of the following grayscale morphological expressions.

(a)* $R_g^D(f) = \left[R_g^E(f^c) \right]^c$.

(b) $R_g^E(f) = \left[R_g^D(f^c) \right]^c$.

9.48 Prove the validity of the following grayscale morphological expressions.

(a)* $(f \ominus nb)^c = (f^c \oplus n\hat{b})$, where $(f \ominus nb)$ indicates n successive erosions, starting with b .

(b) $(f \oplus nb)^c = (f^c \ominus n\hat{b})$.

9.49 Prove the validity of the following grayscale morphological expressions. Recall that $f^c(x, y) = -f(x, y)$ and that $\hat{b}(x, y) = b(-x, -y)$. Assume a symmetric structuring element.

(a)* $O_R^{(n)}(f) = \left[C_R^{(n)}(f^c) \right]^c$.

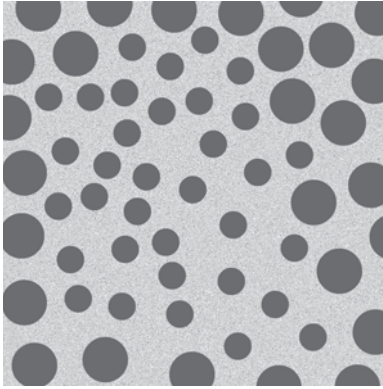
(b) $C_R^{(n)}(f) = \left[O_R^{(n)}(f^c) \right]^c$.

9.50 Consider the image below, which shows a region of small circles enclosed by a region of larger circles.

- (a) Would you expect the method used to generate Fig. 9.45(d) to work with this image also? Explain your reasoning, including any

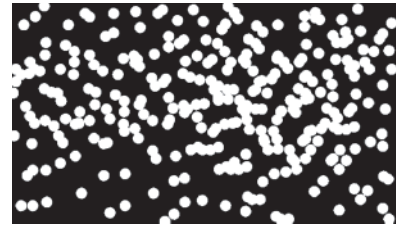
assumptions that you need to make for the method to work.

- (b)* If your answer to (a) is yes, sketch what the boundary will look like.

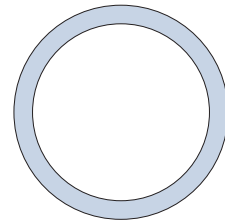


- 9.51** A preprocessing step in an application of microscopy is concerned with the issue of isolating individual round particles from similar particles that overlap in groups of two or more particles (see the following image). Assuming that all particles are of the same size, propose a morphological algorithm that produces three images consisting respectively of:

- (a)* Only particles that have merged with the boundary of the image.
- (b) Only overlapping particles.
- (c) Only nonoverlapping particles.



- 9.52** A high-technology manufacturing plant is awarded a government contract to manufacture high-precision washers of the form shown. The terms of the contract require that the shape of all washers be inspected by an imaging system. In this context, shape inspection refers to deviations from round on the inner and outer edges of the washers. You may assume the following: (1) A “golden” (perfect with respect to the problem) image of an acceptable washer is available; and (2) the imaging and positioning components ultimately used in the system will have an accuracy high enough to allow you to ignore errors due to digitalization and positioning. You are hired as a consultant to help specify the visual inspection part of the system. Propose a solution based on morphological/logical operations.



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